

Assignments in **BOOK 4**

SENIOR MATHS

Kevin Dunne



Quality ready-to-use
resources



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SENIOR
MATHS

BOOK 4

Kevin Dunne



Acknowledgement

I would like to thank my wife Rikki for her invaluable assistance in the preparation of this book.

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I **introduction**

This series has been written to provide comprehensive coverage of selected topics from the senior mathematics syllabus. Book 4 concentrates on coordinate geometry and functions.

Most topics are aimed at the senior syllabus but some would be useful in junior courses. Each worksheet has been carefully graded with the more difficult questions towards the end of each set. Where appropriate, exercises include a worked example to show the most common methods of working. Students should be able to complete the worksheets with minimal help from the teacher. Each topic should have sufficient questions to occupy students for at least a 40-minute lesson.

1	distance formula	5
2	midpoint formula	6
3	gradient formula	7
4	gradient: parallel and perpendicular lines, angles of inclination	8
5	distance, midpoint and gradient review	10
6	introduction to straight lines	11
7	using intercepts to graph straight lines	12
8	general equations: finding the gradient and y-intercept	13
9	point-gradient formula for the equation of a straight line	15
10	two-point formula for the equation of a straight line	16
11	intercept formula and lines parallel to the axes	17
12	finding equations of straight lines: review	19
13	determining whether a point lies on a line	20
14	intersection of two straight lines - simultaneous equations	21
15	parallel and perpendicular lines	22
16	collinear points - using the gradient	23
17	perpendicular distance from a point to a straight line	24
18	equation of a line through the intersection of two straight lines 1	25
19	equation of a line through the intersection of two straight lines 2	26
20	concurrent lines	27
21	linear functions: review 1	28
22	linear functions: review 2	29
23	linear inequalities	30
24	graphs of equations involving absolute values	31
25	function notation	32
26	domain and range	33
27	odd and even functions	34
28	split functions	35
29	intersection of linear and non-linear graphs	36
30	exponential and logarithmic functions	37
31	semi-circles and hyperbolae	38
32	non-linear inequalities	39
33	functions: review 1	40
34	functions: review 2	41
35	examination-style questions 1	42
36	examination-style questions 2	43
37	examination-style questions 3	44
	answers	45

1 distance formula

The distance between two points (x_1, y_1) and (x_2, y_2) is: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

This formula is also used to find the length of an interval joining two points.

e.g.

- 1** Find the distance between the points (4,5) and (8,8).

$$\begin{aligned}d &= \sqrt{(8-4)^2 + (8-5)^2} \\&= \sqrt{4^2 + 3^2} \\&= \sqrt{25} \\&= 5\end{aligned}$$

$\therefore$ The distance between the two points is 5 units.

- 2** Determine the length of the interval joining (3,-5) and (8,-4).

$$\begin{aligned}d &= \sqrt{(8-3)^2 + (-4+5)^2} \\&= \sqrt{5^2 + 1^2} \\&= \sqrt{26}\end{aligned}$$

$\therefore$ The length is exactly $\sqrt{26}$ units long.

Note: Sometimes it is convenient to leave the answer in **exact** form (that is, as a surd). At other times, it is necessary to write the answer as an approximation, to a number of decimal places.

Determine the **exact** distance between the given points.

- | | |
|------------------------|------------------------|
| 1 (1,5) and (4,9) | 2 (6,10) and (0,2) |
| 3 (3,5) and (8,10) | 4 (6,8) and (4,0) |
| 5 (5,12) and (6,7) | 6 (8,4) and (4,1) |
| 7 (3,4) and (-1,-4) | 8 (-6,4) and (5,-2) |
| 9 (-3,-2) and (-5,-4) | 10 (3,-5) and (-4,-6) |
| 11 (5,-3) and (-1,4) | 12 (-6,-4) and (-3,-9) |
| 13 (-7,-4) and (-1,-6) | 14 (3,-5) and (-2,-9) |

Determine (to one decimal place) the length of the interval joining each pair of coordinates.

- | | |
|-----------------------|------------------------|
| 15 (4,3) and (6,8) | 16 (3,5) and (-8,-4) |
| 17 (-4,8) and (-5,-3) | 18 (-6,-3) and (-1,-4) |
| 19 (-1,4) and (3,-4) | 20 (4,-2) and (6,-5) |
| 21 (-5,-7) and (4,-9) | 22 (-7,-1) and (4,-5) |

- 23 Show that the triangle with vertices at A(4,5), B(1,3) and C(2,8) is isosceles.
- 24 Show that the points X(-1,1), Y(3,2) and Z(0,5) are the vertices of an isosceles triangle.
- 25 Use Pythagoras' rule to show that the points A(1,5), B(3,4) and C(7,12) are the vertices of a right-angled triangle.
- 26 Show that the points (-1,1) and (6,8) both lie on the circumference of a circle with its centre at (3,4).
- 27 Show that the points (-1,3), (5,6), (3,-1) and (-3,-4) are the vertices of a parallelogram.
- 28 The points (-1,y) and (4,9) are 13 units apart. Determine the value(s) of y.

The formula for determining the coordinates of the midpoint $M(x,y)$ of the interval joining the points (x_1, y_1) and (x_2, y_2) is:

$$x = \frac{x_1 + x_2}{2} \quad y = \frac{y_1 + y_2}{2}$$

e.g. Determine the coordinates of the midpoint of the interval joining $(6,3)$ to $(-3,9)$.

$$\begin{aligned} x &= \frac{6 + (-3)}{2} & y &= \frac{3 + 9}{2} \\ &= \frac{3}{2} & &= \frac{12}{2} \\ &= 1\frac{1}{2} & &= 6 \end{aligned}$$

$\therefore$ The midpoint of the interval is $(1\frac{1}{2}, 6)$.

Determine the coordinates of the midpoint of the interval joining each pair of coordinates.

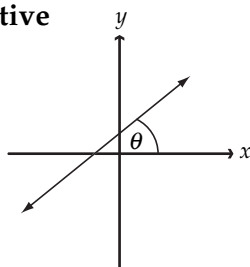
- 1 $(3,5)$ and $(9,3)$
 - 2 $(6,8)$ and $(4,2)$
 - 3 $(3,4)$ and $(5,6)$
 - 4 $(8,1)$ and $(3,2)$
 - 5 $(7,8)$ and $(-3,-4)$
 - 6 $(-3,-4)$ and $(6,-2)$
 - 7 $(0,0)$ and $(7,8)$
 - 8 $(-3,4)$ and $(-3,8)$
 - 9 $(6,-5)$ and $(0,-4)$
 - 10 $(-11,2)$ and $(-4,2)$
 - 11 $(3,0)$ and $(0,5)$
 - 12 $(-3,2)$ and $(4,-1)$
 - 13 $(4,-5)$ and $(-3,-5)$
 - 14 $(-3,-7)$ and $(5,9)$
 - 15 $(-6,-8)$ and $(-3,-7)$
 - 16 $(3,-14)$ and $(7,-20)$
 - 17 $(-15,-7)$ and $(-4,-8)$
 - 18 $(-6,-3)$ and $(-15,-9)$
- 19 The point $(6,7)$ is the midpoint of the interval joining $(4,4)$ to (x,y) . Determine the values of x and y .
 - 20 $(2\frac{1}{2}, 1\frac{1}{2})$ is the midpoint of the line joining (x,y) to $(-3,5)$. Find the values of x and y .
 - 21 The midpoint of the interval joining A to B is $(2\frac{1}{2}, 4\frac{1}{2})$. A has coordinates of $(-5,1)$. Determine the coordinates of B.
 - 22 M lies on the line joining $L(-4,-3)$ to $N(-8,1)$. $LN = 2 \times MN$; determine the coordinates of M.
 - 23 The points $(-1,3)$, $(5,7)$ and $(3,6)$ are the vertices of a triangle. Determine the midpoint of the longest side.
 - 24 The points $(2,6)$, $(4,5)$ and $(8,13)$ are the vertices of a right-angled triangle. Find the midpoint of the hypotenuse.
 - 25 The point $M(8,9)$ is the midpoint of the line LN. The coordinates of L are $(5,6)$. Determine the coordinates of N and then determine the coordinates of O, the midpoint of MN.
 - 26 The coordinates of P and Q are $(-8,3)$ and $(4,-5)$ respectively; determine the coordinates of the point R so that $RQ = \frac{1}{4}PQ$.
 - 27 The line AB is the diameter of a circle, centre $(12, \frac{1}{2})$. The coordinates of A are $(8,5)$; determine the coordinates of B.

3 gradient formula

The gradient (m) of the interval joining the points (x_1, y_1) and (x_2, y_2) is: $m = \frac{y_2 - y_1}{x_2 - x_1}$

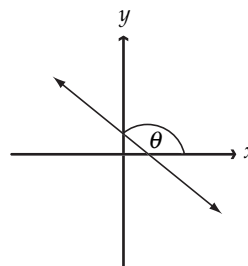
A gradient is either **positive** or **negative**. θ is the **angle of inclination**.

positive



If the gradient is positive, then $0^\circ < \theta < 90^\circ$.
That is, θ is an acute angle.

negative



If the gradient is negative, then $90^\circ < \theta < 180^\circ$.
That is, θ is an obtuse angle.

Note: A line parallel to the x -axis has a slope (gradient) of 0. A line parallel to the y -axis has a slope that is undefined.

e.g. Determine the gradient of the line joining $(8, -5)$ and $(3, 4)$.

$$m = \frac{4 - (-5)}{3 - 8}$$

$$= \frac{9}{-5}$$

$\therefore$ The gradient is $-\frac{9}{5}$.

Plot each pair of points and then give the sign of the gradient of the interval joining them.

- | | |
|-----------------------------|-----------------------------|
| 1 $(3, 4)$ and $(6, 8)$ | 2 $(6, 10)$ and $(5, 1)$ |
| 3 $(-6, -4)$ and $(-3, -5)$ | 4 $(-6, -4)$ and $(-8, -9)$ |
| 5 $(-3, -4)$ and $(4, -2)$ | 6 $(0, 3)$ and $(3, 0)$ |

Determine the gradient of the interval joining each pair of points.

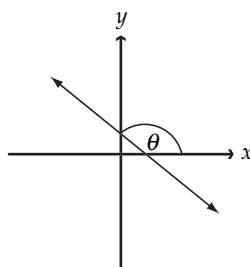
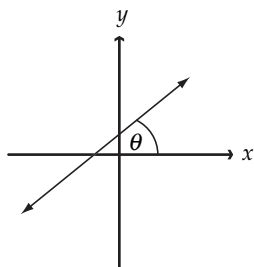
- | | |
|------------------------------|-----------------------------|
| 7 $(8, 5)$ and $(10, 7)$ | 8 $(8, 6)$ and $(1, 7)$ |
| 9 $(-6, -4)$ and $(8, 10)$ | 10 $(-3, -4)$ and $(-5, 9)$ |
| 11 $(8, 10)$ and $(-3, -4)$ | 12 $(6, 10)$ and $(9, -5)$ |
| 13 $(-3, -5)$ and $(-6, 10)$ | 14 $(-8, -5)$ and $(-5, 4)$ |

- 15 The gradient of the interval joining $(5, 4)$ to $(3, y)$ is $\frac{1}{2}$. Determine the value of y .
- 16 Determine the value of x , given that the interval joining $(x, 4)$ to $(3, -2)$ has a gradient of 2.
- 17 The line joining the points $(6, 5)$ and $(x, 3)$ has a gradient of 1. Determine the value of x .
- 18 Show that the angle of inclination of the line joining $(8, 3)$ and $(1, 0)$ is in the range of 0° to 90° .
- 19 Show that the line joining $(-6, 8)$ to $(1, 4)$ makes an angle that is between 90° and 180° with the positive direction of the x -axis.
- 20 Show that the angle of inclination of the line joining $(-8, -7)$ to $(1, 0)$ is acute.
- 21 The angle of inclination of the interval joining $(3, 4)$ and $(-5, -8)$ is obtuse. Is this statement true or false?

The gradient of one line is m_1 and the gradient of a second line is m_2 .

- If the lines are **parallel** then $m_1 = m_2$
- If the lines are **perpendicular** then $m_1 \times m_2 = -1$

The **angle of inclination** (θ) of a straight line is the angle between the straight line and the positive direction of the x -axis.



Gradient of the line = $\tan \theta$

e.g.

- 1** Determine whether the line passing through the points (7,5) and (8,8) is perpendicular or parallel to the line passing through the points (6,4) and (3,5).

$$\begin{aligned} m_1 &= \frac{y_2 - y_1}{x_2 - x_1} & m_2 &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{8 - 5}{8 - 7} & &= \frac{5 - 4}{3 - 6} \\ &= \frac{3}{1} & &= -\frac{1}{3} \end{aligned}$$

$$\begin{aligned} \therefore m_1 \times m_2 &= \frac{3}{1} \times -\frac{1}{3} \\ &= -1 \end{aligned}$$

$\therefore$ The lines are perpendicular.

- 2** Determine the size of the angle of inclination of the line passing through the points (3,4) and (8,10).

$$\begin{aligned} m &= \tan \theta = \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{10 - 4}{8 - 3} \\ &= \frac{6}{5} \end{aligned}$$

$$\therefore \theta = \tan^{-1}\left(\frac{6}{5}\right)$$

$$\therefore \theta = 50^\circ 12'$$

- 1 Give the gradient of the line perpendicular to a line with the gradient $\frac{5}{6}$.
- 2 Give the gradient of the line parallel to a line with the gradient -2 .
- 3 Show that the line passing through (6,5) and (8,9) is parallel to the line passing through (-1,-3) and (4,7).
- 4 Show that the line passing through (3,1) and (4,-1) is parallel to the line passing through (4,1) and (7,-5).
- 5 Show that the line passing through (6,4) and (1,-3) is perpendicular to the line passing through (3,-1) and (-4,4).
- 6 Show that the line passing through (-4,1) and (-2,0) is perpendicular to the line passing through (-3,-5) and (1,3).
- 7 Is the line joining (-4,2) to (1,9) parallel or perpendicular to the line passing through (-1,-5) and (4,2)?



4 gradient: parallel and perpendicular lines, angles of inclination

cont'd

Determine the gradient (in exact form) of a line with the given angle of inclination.

8 30°

9 60°

10 150°

11 135°

Determine the gradient (to one decimal place) of a line with the given angle of inclination.

12 70°

13 40°

14 115°

15 160°

Determine the angle of inclination (to the nearest degree) of a straight line with the given gradient.

16 0.5

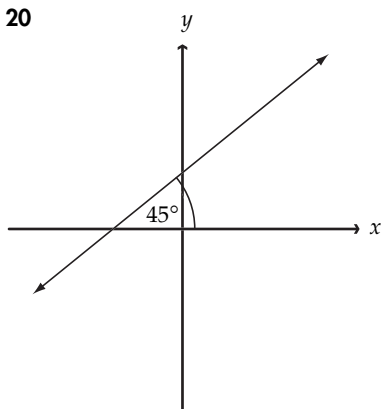
17 1.6

18 -1.3

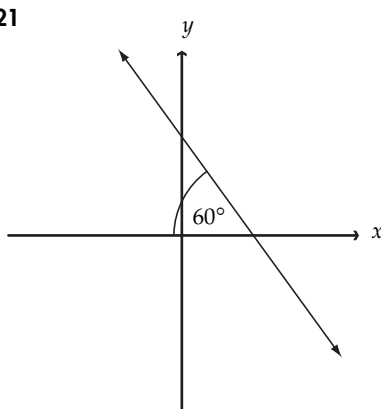
19 -5.8

Determine the gradient of each straight line. Find the exact value for questions 20 and 21 and answer to one decimal place for questions 22–25.

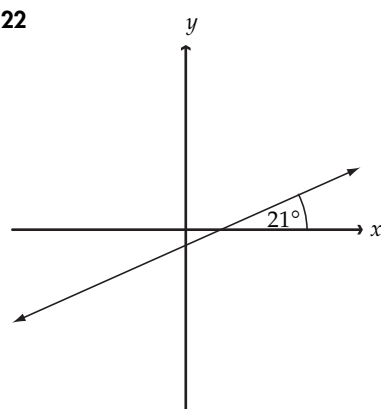
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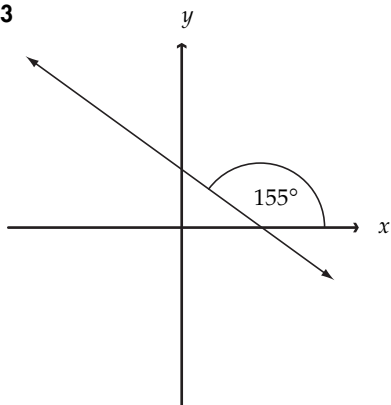
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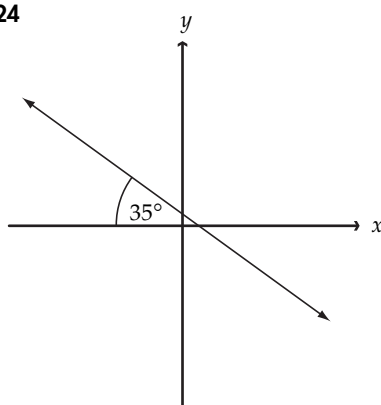
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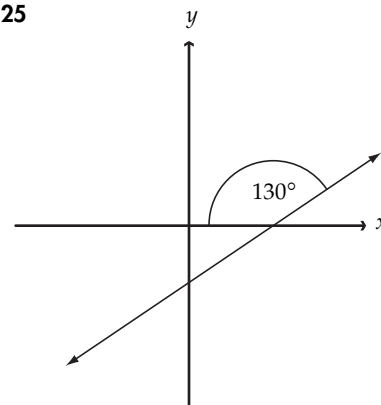
23



24



25



- 26 A straight line passes through the points (4,6) and (8,10). Find the angle the line makes with the positive direction of the x -axis.
- 27 Determine to the nearest degree the angle of inclination of the line joining the points (-4,-1) and (1,5).
- 28 Determine, to the nearest degree, the angle of inclination of the line joining the points (-4,6) and (1,1).
- 29 The line joining the points (3,4) and (6,10) is parallel to the line joining (-1,-1) to (2, y). Determine the value of y .
- 30 The line joining P and Q is perpendicular to the line joining R and S. The coordinates of P, Q, R and S are (3,1), (7,10), (1,6) and (x ,2), respectively. Determine the value of x .

distance formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

gradient formulae

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta \quad \text{where } \theta \text{ is the angle of inclination}$$

midpoint formula

$$x = \frac{x_1 + x_2}{2} \quad y = \frac{y_1 + y_2}{2}$$

$$\text{For parallel lines: } m_1 = m_2$$

$$\text{For perpendicular lines: } m_1 \times m_2 = -1$$

- 1 A line passes through the points A(3,5) and B(7,10).
 - a Calculate the length of AB.
 - b Determine the coordinates of the midpoint of AB.
 - c Find the gradient of AB.
 - d What is the angle of inclination of the line AB?
 - e Point C has coordinates (8,1). Show that AC is perpendicular to AB.
- 2 A line passes through the points P(-3,1) and Q(-5,4).
 - a Calculate the length of PQ.
 - b Determine the coordinates of the midpoint of PQ.
 - c Find the gradient of PQ.
 - d What is the angle of inclination of the line PQ?
 - e Show that the line joining (5,4) and (3,7) is parallel to PQ.
- 3 The midpoint of the line joining (5,8) to (-1,y) is $(2, 6\frac{1}{2})$. Find the value of y.
- 4 Show that the points (1,4), (-2,2) and (3,1) form a right-angled triangle. (Hint: Use gradients.)
- 5 In a parallelogram, both pairs of opposite sides are parallel. Show that the quadrilateral formed by joining A(-5,0), B(1,5), C(3,1) and D(-3,-4) is a parallelogram.
- 6 The points P(-3,5), Q(-3,-1), R(3,-1) and S form a square. Give the coordinates of S. Hence show that the diagonals are equal, bisect each other and are perpendicular to each other.
- 7 The points A(2,10), B(7,5) and C(1,3) form a triangle. Show that $\triangle ABC$ is isosceles. Point D is the midpoint of BC. Determine the coordinates of D and show that AD is perpendicular to BC.
- 8 Determine the angle made by the line through the points (-4,-1) and (5,4) with the positive direction of the x-axis.
- 9 The angle of inclination of the line joining the point (x,5) to (-1,3) is 135° . Determine the value of x.

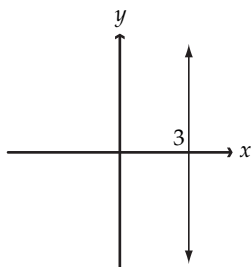
6 introduction to straight lines

It can be helpful to classify a straight line by its equation. Every straight line can be classified as one of three types.

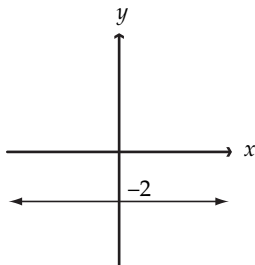
Parallel to an axis

e.g.

1 $x = 3$



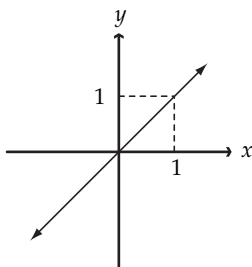
2 $y = -2$



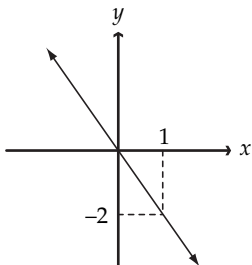
Through the origin

e.g.

1 $y = x$



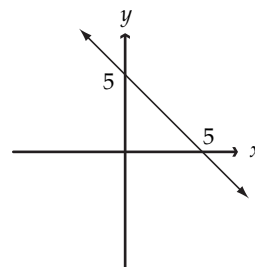
2 $y = -2x$



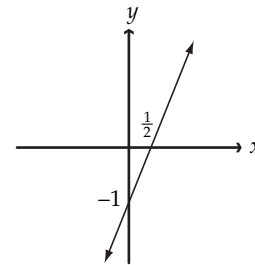
Cutting both axes

e.g.

1 $x + y = 5$



2 $y = 2x - 1$



Give the class of straight line **without** sketching the equation.

1 $y = 3$

2 $y = 3x$

3 $x = 4$

4 $x = -3$

5 $y = 1 - 3x$

6 $6 + x + y = 0$

7 $2x + y = 5$

8 $y = 3x - 1$

9 $x + 4 = 0$

10 $y - 3 = 0$

11 $2x - y = 0$

12 $3x + 4y = 2$

13 $3y = 2x + 5$

14 $3x = 8y - 5$

15 $3y - 2x = 0$

16 $4x + 5y = 0$

17 Give the equations of the x -axis and the y -axis.

For each of the following questions, draw neat sketches of the three lines on the same set of axes.

18 a $y = 4$

19 a $y = -x$

20 a $y = \frac{1}{2}x$

21 a $y = 2$

b $x + y = 1$

b $x = -1$

b $y = 2x + 2$

b $2y = x + 2$

c $y = 2x$

c $x - y = 3$

c $y + 3 = 0$

c $x = 3y$

22 a $x + y + 3 = 0$

23 a $x = 4y$

24 a $x + 4 = 0$

25 a $2x - 3y = 6$

b $y + \frac{1}{3}x = 0$

b $2x - 3 = 0$

b $x + 2y = 0$

b $y + 4 = 0$

c $x - 5 = 0$

c $x - 2y = 4$

c $2y = 3x - 6$

c $2x + 3y = 0$

A quick method for graphing straight lines is to determine the x - and y -intercepts.

Note: When a straight line cuts the x -axis, the y coordinate is 0.

When a straight line cuts the y -axis, the x coordinate is 0.

e.g. Use the intercept method to sketch the line with the equation $2x + 3y = 6$.

To find the x -intercept, let $y = 0$.

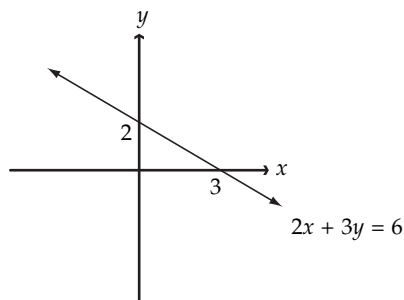
$$\text{That is: } 2x = 6$$

$$\therefore x = 3$$

To find the y -intercept, let $x = 0$.

$$\text{That is: } 3y = 6$$

$$\therefore y = 2$$



Note: This method will only work for equations that have three separate terms; that is, lines that cut **both** axes.

Give the x - and y -intercepts for each equation.

1 $x + y = 4$

3 $2x + y = 6$

5 $3x - 2y = 9$

7 $2x - 4y = 10$

9 $2x + 5y = 4$

11 $y - 3x = 6$

13 $x - 2y + 3 = 0$

15 $2y = 3x - 1$

17 $4 + 2y = 5x$

19 $4x = 3y - 5$

2 $x + y = -6$

4 $3x + 2y = 3$

6 $3x - y = 7$

8 $3y + 2x = 7$

10 $y - x = 4$

12 $3x + 2y - 12 = 0$

14 $0 = 2y - 4x + 5$

16 $2y + 3x = 3$

18 $2 - 5x = 7y$

20 $5y - 4x - 6 = 0$

For each equation, determine the x - and y -intercepts and then draw a neat sketch of the graph.

21 $x + y = 3$

23 $2x + y = 8$

25 $2x - 5y = 10$

27 $3y - 2x = 9$

29 $4x + 5y = -10$

31 $5x + 4y = 6$

33 $5x + 7y = 0$

22 $x - y = 4$

24 $3y + 2x = 6$

26 $5y - x = 5$

28 $2x + 3y = 5$

30 $2x - 3y = 4$

32 $3x - 5y + 4 = 0$

34 $6x - 2y - 7 = 0$

The general equation of a straight line is $ax + by + c = 0$.

Transpose to make y the subject: $y = -\frac{a}{b}x - \frac{c}{b}$

where $-\frac{a}{b}$ is the gradient and $-\frac{c}{b}$ is the y -intercept.

Another way of writing this is: $y = mx + b$ where m is the gradient and b is the y -intercept.

e.g.

- 1** Give the gradient and y -intercept of the straight line with the equation $2x + 3y + 4 = 0$.

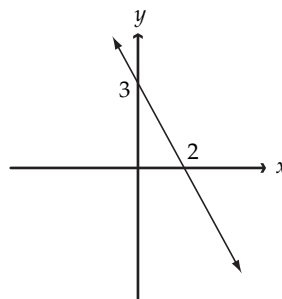
$$2x + 3y + 4 = 0$$

$$3y = -2x - 4$$

$$\therefore y = -\frac{2}{3}x - \frac{4}{3}$$

Thus the gradient is $-\frac{2}{3}$ and the y -intercept is $-\frac{4}{3}$.

- 2** The graph of a straight line is given below. Write the equation in general form.



$m = -\frac{3}{2}$ (as graph slopes to the left) and the y -intercept is 3

The equation is $y = -\frac{3}{2}x + 3$

$$2y = -3x + 6$$

$$\therefore 3x + 2y - 6 = 0$$

For each linear equation, give the gradient and the y -intercept.

1 $2x + y + 3 = 0$

3 $y - 2x + 5 = 0$

5 $2x + 3y + 5 = 0$

7 $4x + 3y = 2$

9 $3x = 2y + 5$

11 $3x + 4y = 0$

2 $3x + y - 4 = 0$

4 $2x + 2y - 3 = 0$

6 $3y - 2x + 4 = 0$

8 $6x + 2y = 5$

10 $7x = 3 - 4y$

12 $2x - 3y = 0$

For each straight line with gradient m and y -intercept b , write the linear equation in general form.

13 $m = 3, b = 4$

15 $m = 5, b = 2$

17 $m = \frac{2}{3}, b = 4$

19 $m = -\frac{3}{4}, b = 5$

21 $m = -\frac{2}{3}, b = -\frac{2}{3}$

14 $m = 2, b = 1$

16 $m = -3, b = -1$

18 $m = \frac{1}{2}, b = 3$

20 $m = -\frac{2}{5}, b = -3$

22 $m = \frac{1}{2}, b = \frac{1}{3}$

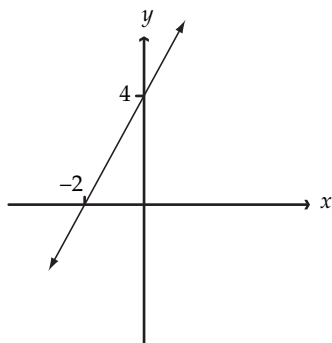


8 general equations: finding the gradient and y-intercept

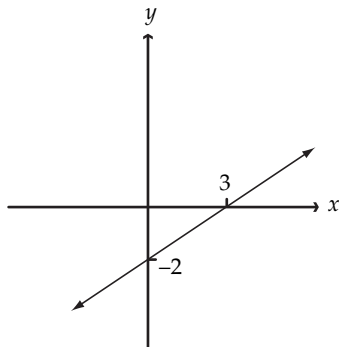
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For each sketch, give the equation of the straight line. (Hint: Some lines have a negative slope.)

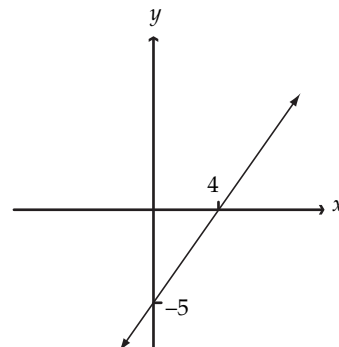
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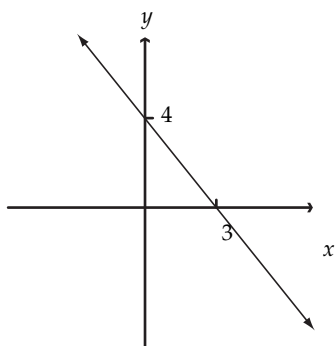
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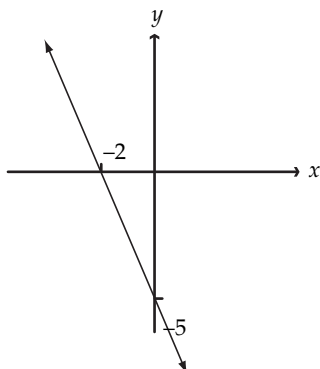
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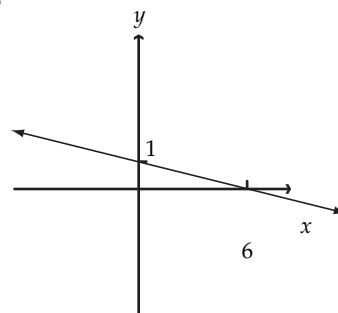
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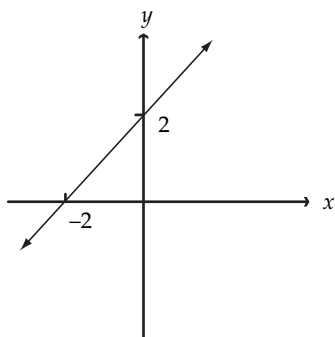
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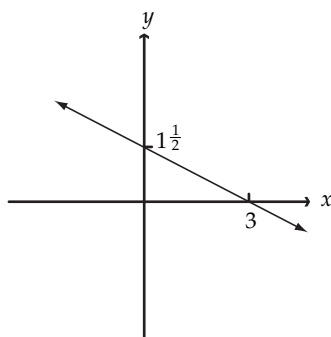
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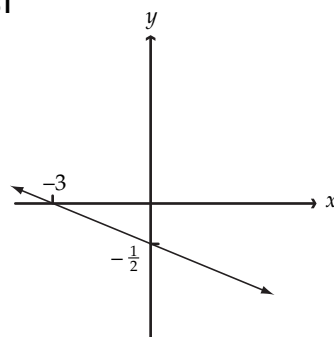
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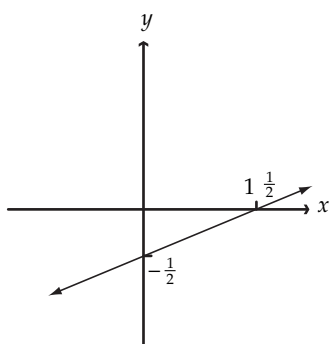
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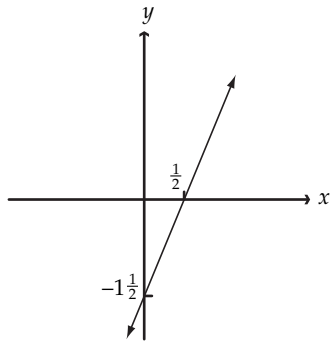
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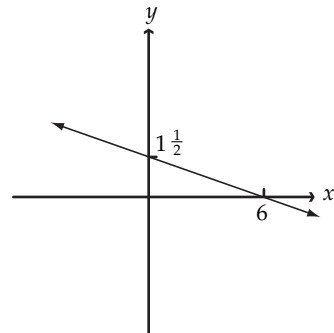
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9 point-gradient formula for the equation of a straight line

The equation of a straight line, given the gradient (m) and a point (x_1, y_1) it passes through, is:

$$m = \frac{y - y_1}{x - x_1} \quad \text{or} \quad y - y_1 = m(x - x_1).$$

The gradient $m = \tan \theta$ where θ is the angle of inclination.

e.g. Determine the equation of a straight line that has a gradient of 4 and passes through the point $(3, -4)$.

$$y - y_1 = m(x - x_1)$$

$$y + 4 = 4(x - 3)$$

$$y + 4 = 4x - 12$$

$$\therefore 4x - y - 16 = 0$$

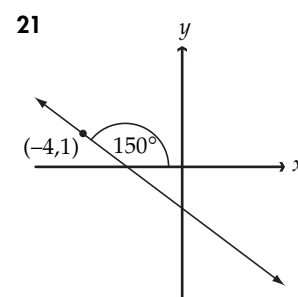
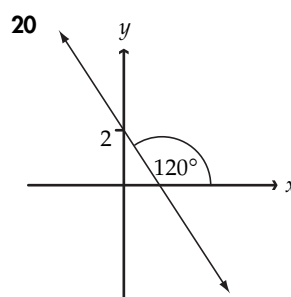
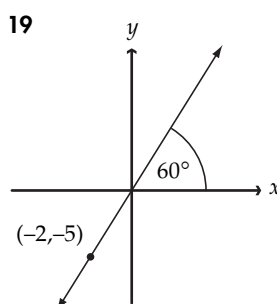
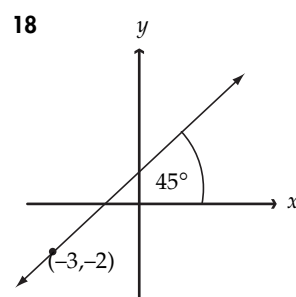
Determine the equation given that the straight line has:

- | | | |
|--|--|---|
| 1 a gradient of 3 and passes through the point $(3, 1)$ | 2 a gradient of 4 and passes through the point $(4, 3)$ | 3 a gradient of -1 and passes through the point $(1, -1)$ |
| 4 a gradient of -4 and passes through the point $(-2, 7)$ | 5 a gradient of $\frac{3}{4}$ and passes through the point $(5, 2)$ | 6 a gradient of $\frac{2}{3}$ and passes through the point $(-3, -4)$ |
| 7 a gradient of $-\frac{1}{3}$ and passes through the point $(-3, -1)$ | 8 a gradient of $1\frac{1}{2}$ and passes through the point $(4, 3)$ | 9 a gradient of $-2\frac{1}{2}$ and passes through the point $(-6, -4)$ |
- 10 A line passing through the point $(1, 6)$ has an angle of inclination of θ . Determine the equation of the line given that $\tan \theta = 2$. (Hint: gradient = $\tan \theta$)
- 11 A straight line passes through the point $(-3, 2)$ and has an angle of inclination of θ , where $\tan \theta = 1\frac{1}{2}$. Give the equation of the line in general form.

Find the equation of each line, given that:

- | | |
|--|--|
| 12 $\tan \theta = 2$ and the line passes through the point $(3, 1)$. | 13 $\tan \theta = \frac{1}{2}$ and the line passes through the point $(1, -3)$. |
| 14 $\tan \theta = \frac{3}{4}$ and the line passes through the point $(3, -5)$. | 15 $\tan \theta = -2\frac{1}{3}$ and the line passes through the point $(3, -5)$. |
| 16 $\tan \theta = -3$ and the line passes through the point $(-2, 1)$. | 17 $\tan \theta = -\frac{2}{3}$ and the line passes through the point $(3, -5)$. |

Determine the equation of each straight line, leaving in surd form where necessary.



The equation of a straight line passing through the points (x_1, y_1) and (x_2, y_2) is:

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

e.g. Determine the equation of the straight line that passes through the points $(3, 4)$ and $(6, 9)$.

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y - 4}{x - 3} = \frac{9 - 4}{6 - 3}$$

$$\frac{y - 4}{x - 3} = \frac{5}{3}$$

$$3(y - 4) = 5(x - 3)$$

$$3y - 12 = 5x - 15$$

$$\therefore 5x - 3y - 3 = 0$$

Express in general form the equation of the straight line passing through each pair of points.

1 $(1, 4)$ and $(3, 8)$

2 $(2, 6)$ and $(5, 10)$

3 $(2, 8)$ and $(-1, 4)$

4 $(3, 2)$ and $(-1, -3)$

5 $(4, -3)$ and $(2, 4)$

6 $(-3, -2)$ and $(3, 7)$

7 $(-3, -5)$ and $(-1, -4)$

8 $(-2, -4)$ and $(-5, -2)$

9 $(6, 3)$ and $(0, 4)$

10 $(3, 0)$ and $(-2, -8)$

11 $(4, 1)$ and $(3, -4)$

12 $(6, 7)$ and $(9, 10)$

13 $(-3, -4)$ and $(-6, -8)$

14 $(3, 5)$ and $(-6, -4)$

15 $(8, 0)$ and $(-1, 4)$

16 $(5, 11)$ and $(1, 10)$

17 The points $(8, 1)$ and $(6, 4)$ lie on a straight line. Determine its equation.

18 A straight line passes through the points $(-1, 2)$ and $(-3, 5)$. Find its equation.

19 A straight line passes through the points $(3, -1)$ and $(-4, -3)$. Give its equation in gradient/intercept form.

20 Find the equation of the straight line joining $(-1, 4)$ to $(3, -7)$ and give its y -intercept.

21 Determine the equation of the straight line joining $(-2, 1)$ and $(3, 4)$. Give the coordinates of the point where the line cuts the y -axis.

22 Determine the equation of the straight line passing through the points $(-2, -4)$ and $(2, 4)$. Which part of the equation tells you that the line passes through the origin?

23 Show that the line joining the points $(3, -9)$ and $(-1, 3)$ passes through the origin.

The equation of a straight line, given the x -intercept a and the y -intercept b , is: $\frac{x}{a} + \frac{y}{b} = 1$

The equation of a line passing through the point (x_1, y_1) and:

- parallel to the x -axis is given by: $y = y_1$
- parallel to the y -axis is given by: $x = x_1$

e.g.

- 1** Find the equation of the straight line with an x -intercept of -2 and a y -intercept of 3 .

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{x}{-2} + \frac{y}{3} = 1$$

(multiplying throughout by -6)

$$3x - 2y = -6$$

$$\therefore 3x - 2y + 6 = 0$$

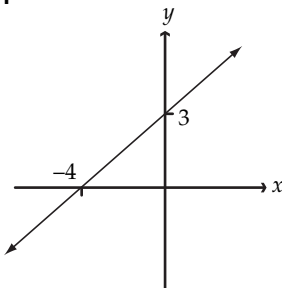
- 2** A line is parallel to the y -axis. Write its equation, given that it also passes through the point $(-2, 1)$.

$$x = x_1$$

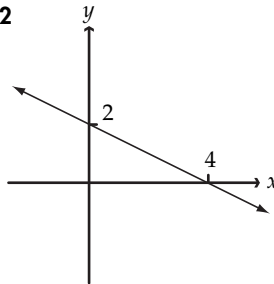
$$\therefore x = 1$$

Give the equation of each straight line.

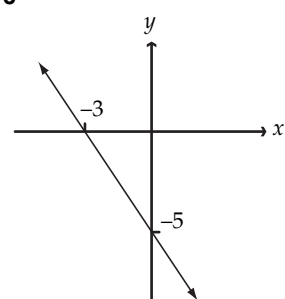
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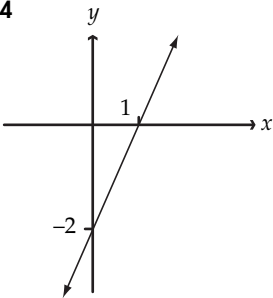
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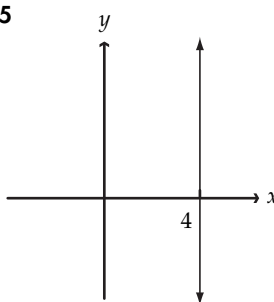
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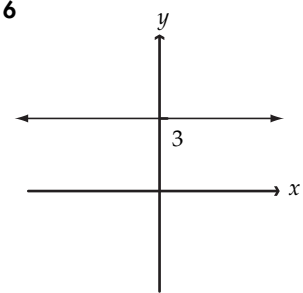
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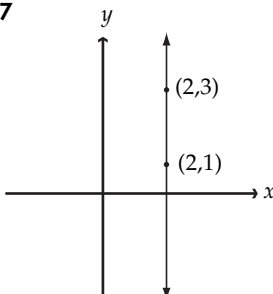
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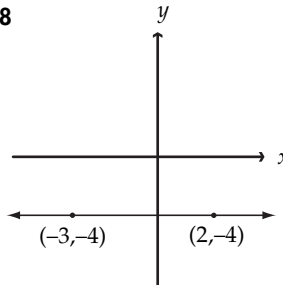
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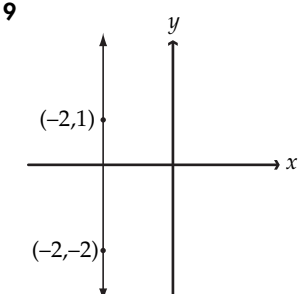
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8



9



- 10 Determine the equation of the straight line that has x - and y -intercepts of 3 and -1 , respectively.
- 11 A straight line has an x -intercept of -3 and a y -intercept of 5. Write its equation.
- 12 The x - and y - intercepts of a straight line are -2 and -3 respectively. Give its equation in general form.
- 13 Given that the x - and y - intercepts of a straight line are $1\frac{1}{2}$ and 2 respectively, determine the equation in general form.
- 14 A straight line is parallel to the x -axis and passes through the point $(-1, -3)$. Give its equation.
- 15 A straight line is parallel to the x -axis. It also passes through the point $(-1, 4)$. Write its equation.
- 16 A straight line is parallel to the y -axis. Given that it passes through the point $(-3, -4)$, write its equation.
- 17 A straight line passes through the point $(-4, -4)$ and is parallel to the y -axis. Write its equation.
- 18 A straight line is always 3 units below the x -axis. Write its equation.
- 19 A line is 4 units above the x -axis. What is its equation?
- 20 A line is 4 units to the right of the y -axis. What is its equation?
- 21 A straight line has an x -intercept of 2. The y -intercept is double the x -intercept. Write its equation.
- 22 The x -intercept of a straight line is $\frac{1}{2}$. Determine the equation of the line given that the y -intercept is 5.
- 23 A straight line is drawn so that it is always 3 units from the x -axis. Give the possible equations of the straight line.
- 24 A straight line is parallel to the y -axis. Given that it is 3 units to the right of the point $(2, 3)$, write its equation.
- 25 A straight line is parallel to the x -axis. Given that it is 3 units below the point $(3, 1)$, write its equation.
- 26 A straight line is drawn so that it is parallel to the x -axis. Another line is drawn parallel to the y -axis. Given that both lines pass through the point $(3, -1)$, write the equation of each line.
- 27 The x -intercept of a straight line is 4. The y -intercept is 3 units from the origin. Give the possible equation(s).

Find the equation of the straight line that passes through each pair of points.

- | | |
|----------------------------|---------------------------|
| 28 $(0, 5)$ and $(-4, 0)$ | 29 $(0, -1)$ and $(3, 0)$ |
| 30 $(0, 4)$ and $(7, 0)$ | 31 $(3, 4)$ and $(-1, 4)$ |
| 32 $(-2, 4)$ and $(-2, 5)$ | 33 $(3, 5)$ and $(3, -4)$ |

12 finding equations of straight lines: review

You may find the following formulae helpful.

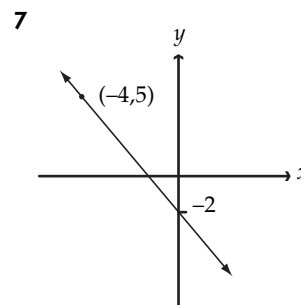
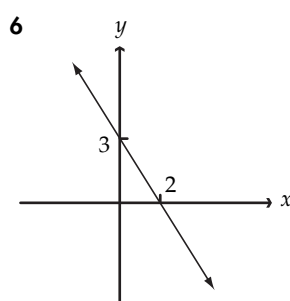
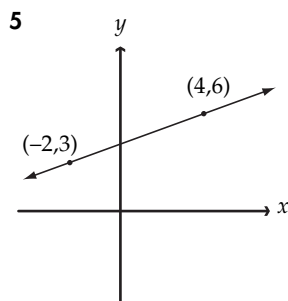
$$y - y_1 = m(x - x_1) \quad \text{or} \quad m = \frac{y - y_1}{x - x_1} \quad y = mx + b$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1} \quad \frac{x}{a} + \frac{y}{b} = 1$$

Note: The two-point formula and the intercepts formula are useful to know. However, as all equations of straight lines can be determined by the first two formulae, they are not essential.

Give the equation of the straight line that passes through each pair of points.

- 1 (1,4) and (-3,-2) 2 (3,8) and (-1,-3) 3 (1,2) and (-4,5) 4 (5,6) and (-1,7)



- 8 Determine the equation of the straight line that passes through the point (3,5) and has a gradient of 5.
- 9 A straight line with a gradient of -4 passes through the point (3,-2). Find its equation.
- 10 A straight line passes through the point (4,3) and has a gradient of $\frac{2}{5}$. Find its equation.
- 11 Determine the equation of a straight line passing through the points (1,4) and (3,5).
- 12 A straight line passes through (5,4) and (-1,-1). Determine its equation.
- 13 Given that a straight line passes through (3,-2) and (-1,4), determine its equation.
- 14 A straight line has an x-intercept of 4 and a y-intercept of -1. Write its equation in general form.
- 15 A straight line passes through (0,4) and (-3,0). Write its equation in general form.
- 16 Given that the x- and y-intercepts of a straight line are -2 and 5 respectively, determine the equation of the line.
- 17 A straight line passes through (0,3). Given that it has a gradient of $-\frac{2}{3}$, find its equation.
- 18 Show that the line joining (-2,-6) and (3,9) passes through the origin.
- 19 Determine the equation of the line passing through the origin and the point (-3,-5).
- 20 Write the equation of the straight line that is parallel to the x-axis and passes through the point (3,5).
- 21 Find the equation of the straight line that is parallel to the y-axis and also passes through the point $(3\frac{1}{2}, 6)$.
- 22 The point $(7, -2\frac{1}{2})$ lies on a line that is parallel to the y-axis. Write the equation of the line.
- 23 A straight line passes through the points (4,7) and (-1,7). Find the equation of the line.
- 24 The points (-3,-4) and (-3,-5) lie on a straight line. Determine its equation.
- 25 The points A(-3,4), B(4,4), C(4,-2) and D(-3,-2) form a rectangle. Write the equations of the lines AB and BD.

To determine if a point lies on a straight line, substitute the coordinates of that point into the equation. If the coordinates satisfy the equation, then the point is on that line.

Collinear points are points that lie on the same straight line.

e.g.

- 1** Show that the point (4,5) lies on the straight line with the equation $2x + y - 13 = 0$.

Substitute $x = 4$, $y = 5$ into the left hand side of the equation.

$$\text{L.H.S.} = 2 \times 4 + 5 - 13$$

$$= 0$$

$$= \text{R.H.S}$$

$$\therefore (4,5) \text{ lies on the line } 2x + y - 13 = 0$$

- 2** Show that the points (2,-3), (4,-2) and (-2,-5) are collinear.

The equation of the line joining (2,-3) to (4,-2) is:

$$\frac{y+3}{x-2} = \frac{-2+3}{4-2}$$

$$\frac{y+3}{x-2} = \frac{1}{2}$$

$$\therefore 2y + 6 = x - 2$$

$$\therefore x - 2y = 8$$

Substitute (-2,-5) into $x - 2y = 8$

$$\text{L.H.S.} = -2 - 2 \times -5$$

$$= 8 = \text{R.H.S.}$$

$\therefore$ The points are collinear.

-
- 1 Show that the point (3,4) lies on the line $3x - y - 5 = 0$.
 - 2 Show that the point (-1,-2) lies on the line $2x - y = 0$.
 - 3 Show that the point (2,-1) lies on the line $3x + y = 5$.
 - 4 Show that the point (6,1) lies on the line $x - 3y = 3$.
 - 5 Does the point (5,-1) lie on the line $x + 3y = 2$?
 - 6 Show that (3,1) does not lie on the line with the equation $3x - 2y + 7 = 0$.
 - 7 Determine whether the point (-3,-1) lies on the straight line $3x - 2y + 7 = 0$.
 - 8 Show that the point (-3,-6) lies on the line $2y = 2x - 7$.
 - 9 The point (3,4) lies on the line $ax - 2y = -1$. Determine the value of a .
 - 10 The point $(x_1, 7)$ lies on the line $3x + 2y = -7$. What is the value of x_1 ?
 - 11 The point (4,-3) lies on the line $y = 2x + c$. Determine the value of c .
 - 12 Show that the points (2,5), (0,1) and (-1,-1) all lie on the straight line $y = 2x + 1$.
 - 13 Which of the points (-1,-3), (0,-2) and (1,3) lie on the line $2x - y - 1 = 0$?
 - 14 Determine the equation of the straight line joining (3,4) to (6,7) and hence show that the points (3,4), (6,7) and (9,10) are collinear.
 - 15 Show that the points (2,-1), (-2,-9) and (3,1) are collinear.
 - 16 Show that the points (1,0), (3,4) and (-3,-8) are collinear.
 - 17 Show that the points (2,1), (-2,-5) and (-4,-8) are collinear.
 - 18 The points (-1,2), (1,6) and (h,3) are collinear. Determine the value of h .

14 intersection of two straight lines – simultaneous equations

To determine the coordinates of the point of intersection of 2 straight lines, solve the equations **simultaneously**.

e.g. Find the coordinates of the point of intersection of the straight lines with equations $x - y = 4$ and $2x + 3y = 13$.

$$x - y = 4 \quad (1)$$

$$2x + 3y = 13 \quad (2)$$

$$(1) \times 2$$

$$2x - 2y = 8 \quad (3)$$

$$(2) - (3)$$

$$5y = 5$$

$$\therefore y = 1$$

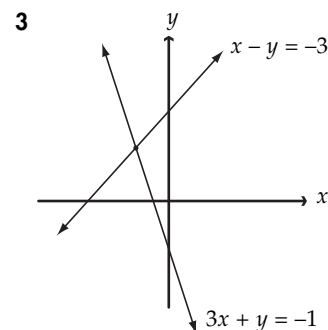
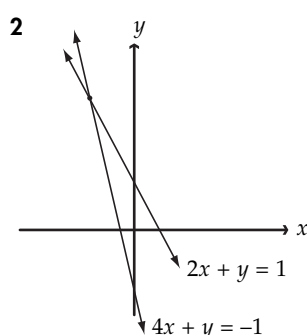
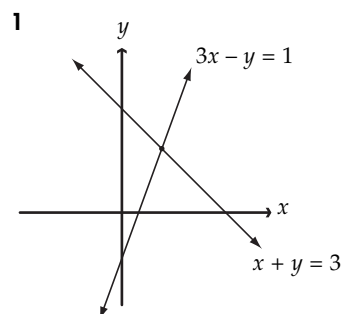
Substitute $y = 1$ into (1)

$$x - 1 = 4$$

$$\therefore x = 5$$

$\therefore$ The point of intersection is (5,1)

Determine the coordinates of each point of intersection.



Determine the coordinates of the point of intersection of each pair of straight lines.

4 $x + y = 3$ and $x - y = 5$

5 $x + y = 1$ and $-x + y = 5$

6 $x - y = 3$ and $y = 5 - x$

7 $x - y = 5$ and $x + y = 8$

8 $x - 2y = -1$ and $y = 5 - x$

9 $2x + y = 4$ and $-2x + y = 2$

10 $x + 2y = 7$ and $3x + 4y = 15$

11 $2x - y = -5$ and $5x + 2y = 1$

11 $3x + 2y = 11$ and $4x + 3y = 16$

13 $4x + 3y = -10$ and $3x - 5y = 7$

14 The x -axis and the lines $2x + y = 5$ and $x - 3y = -1$ form a triangle. Determine the coordinates of the vertices of the triangle.

15 The lines $y = 2x$, $2x + y = 0$ and $2x - 3y = -8$ form a triangle. Determine the coordinates of the vertices of the triangle.

16 Show that the point of intersection of the lines $4x + y = 1$ and $x - y = 4$ also lies on the line $3x + 2y = -3$.

17 Show that the point of intersection of the lines $2x - 3y = -9$ and $3x - 5y = -14$ also lies on the line $7x + 2y = -19$.

15 parallel and perpendicular lines

If two straight lines have gradients m_1 and m_2 , the lines will be:

- **parallel** if $m_1 = m_2$
- **perpendicular** if $m_1 \times m_2 = -1$

To find the gradient of a line for a given equation, write the equation in the form $y = mx + b$, where m is the gradient.

e.g.

- 1** Determine whether the line joining $(-1, -3)$ and $(2, -5)$ and the line joining $(3, 4)$ and $(5, 7)$ are parallel, perpendicular or neither.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m_1 = \frac{-5 - (-3)}{2 - (-1)} = -\frac{2}{3}$$

$$m_2 = \frac{7 - 4}{5 - 3} = \frac{3}{2}$$

$$m_1 \times m_2 = -1$$

$\therefore$ The lines are perpendicular.

- 2** Show that the lines with the equations $3x - 2y + 4 = 0$ and $4y = 6x + 5$ are parallel.

Rewrite the equations in gradient-intercept form:

$$y = \frac{3}{2}x + 2$$

$$\therefore m_1 = \frac{3}{2}$$

$$y = \frac{3}{2}x + \frac{5}{4}$$

$$\therefore m_2 = \frac{3}{2}$$

$$m_1 = m_2$$

$\therefore$ The lines are parallel.

Determine whether the two lines joining each pair of coordinates are parallel, perpendicular or neither.

1 $(3, 4)$ to $(8, 9)$ and $(6, 7)$ to $(-1, 0)$

2 $(2, 5)$ to $(3, 7)$ and $(5, 8)$ to $(10, 12)$

3 $(2, 3)$ to $(5, 7)$ and $(1, 4)$ to $(-3, 7)$

4 $(-3, -4)$ to $(0, 1)$ and $(-2, -3)$ to $(1, 0)$

Determine whether the straight lines represented by each pair of equations are parallel, perpendicular or neither.

5 $2x + 3y = 5$ and $2x - 3y = 4$

6 $x - 2y + 1 = 0$ and $2x - y - 3 = 0$

7 $x + 3y = 5$ and $6 - x - 3y = 0$

8 $\frac{1}{2}x + 2y - 1 = 0$ and $3 - x - 4y = 0$

9 $4y = 3x - 8$ and $8x = 3 - 6y$

10 $2x - 4y = 5$ and $4x - 2y = 2$

11 $2y - x - 3 = 0$ and $3x - 6y + 1 = 0$

12 $3x - 2y + 6 = 0$ and $6x - 1 = 4y$

13 Show that the line joining $(3, 4)$ to $(1, 1)$ is parallel to the line $3x - 2y = 5$.

14 Show that the line joining $(-1, 5)$ and $(-5, 2)$ is perpendicular to $4x + 3y = 5$.

15 The line joining $(1, 1)$ and $(h, 8)$ is parallel to $7x - y + 4 = 0$. Determine the value of h .

16 The line joining $(6, 4)$ to $(3, k)$ is perpendicular to $3x - 5y = 4$. Determine the value of k .

17 Show that the line joining the points $(-1, 5)$, $(6, 0)$ and $(1, -7)$ form a right-angled triangle.

18 Show that the diagonals of the quadrilateral formed by joining the points $(-3, 6)$, $(5, 6)$, $(-3, -2)$ and $(5, -2)$ bisect each other at right angles.

19 The lines with equations $x - y = 1$, $3x + 4y = 12$, $2y = x + 6$ and $8y = -6x - 4$ form a quadrilateral. Show that the quadrilateral is a trapezium.

16 collinear points – using the gradient

Collinear points are points that lie on the same straight line.

Two methods for showing that points are collinear are:

- 1 Determine the equation of the straight line joining two points and show that the third point satisfies the equation (see Topic 13).
- 2 Determine the gradient of the line joining the first two points and then the second and third points.

e.g. Show that the points (3,4), (6,8) and (−3,−4) are collinear.

$$\text{Using the gradient formula } m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\begin{aligned} \text{Gradient of line joining (3,4) and (6,8)} \quad m_1 &= \frac{8-4}{6-3} \\ &= \frac{4}{3} \end{aligned}$$

$$\begin{aligned} \text{Gradient of line joining (6,8) and (−3,−4)} \quad m_2 &= \frac{-4-8}{-3-6} \\ &= \frac{-12}{-9} \\ &= \frac{4}{3} \end{aligned}$$

∴ The three points are collinear.

Show that the points are collinear.

- | | |
|--|--|
| 1 (1,4), (3,2) and (−1,6) | 2 (−1,−3), (0,−2) and (4,2) |
| 3 (−3,10), (1,6) and (4,3) | 4 (−2,1), (4,−5) and (−4,3) |
| 5 (1,3), (3,−1) and (0,5) | 6 (2,1), (6,−1) and (−2,3) |
| 7 (8,1), (−1,−2) and (17,4) | 8 (1,4), (3,−2) and (−2,13) |
| 9 (3,4), (1,6) and (−1,8) | 10 (1,8), (−1,10) and (6,3) |
| 11 (6,−1), (3,−4) and (8,1) | 12 (1,4), (−1,2) and (3,6) |
| 13 (1,2), (−1,6) and (4,−4) | 14 (3,2), (−3,6) and (15,−6) |
| 15 (2,1), (5,2) and (−7,−2) | 16 (2,3), (−6,−9) and (1,1 $\frac{1}{2}$) |
| 17 (1 $\frac{1}{2}$,1), (−3,4) and (1,1 $\frac{1}{3}$) | 18 (1,0), (4,4) and (1 $\frac{1}{2}$,−2 $\frac{2}{3}$) |
-
- 19 Given that (7,1), (5,−1) and (h,−5) are collinear points, determine the value of h.
 - 20 Given that (3,4), (−1,12) and (2,k) are collinear points, determine the value of k.
 - 21 Given that (3,2), (k+1,4) and (12,−1) are collinear points, determine the value of k.
 - 22 Given that (1 $\frac{1}{2}$,0), (2 $\frac{1}{2}$,2) and (h+3,1) are collinear points, find the value of h.
 - 23 Given that (−4,1), (12,−1) and (3k+1,2) are collinear, find the value of k.

17 perpendicular distance from a point to a straight line

The formula for finding the perpendicular distance from the point (x_1, y_1) to the straight line $ax + by + c = 0$ is: $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

Note: By leaving out the absolute value sign, you can use this formula to determine if 2 points lie on the same side or on opposite sides of a straight line.

- If the signs of d are the **same**, the points lie on the **same** side.
- If the signs of d are **different**, the points lie on **opposite** sides of the line.

e.g.

- 1** Find the shortest distance from the point $(5,6)$ to the straight line $3x + 4y = 4$.

$$\begin{aligned} d &= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \\ &= \frac{|3 \times 5 + 4 \times 6 - 4|}{\sqrt{3^2 + 4^2}} \\ &= \frac{35}{5} \\ &= 7 \end{aligned}$$

$\therefore$ The distance is 7 units.

- 2** Show that the points $(-1,1)$ and $(3,2)$ are on opposite sides of the straight line with the equation $3y + 4x - 4 = 0$.

$$\begin{aligned} d &= \frac{3 \times -1 + 4 \times 1 - 4}{\sqrt{3^2 + 4^2}} & d &= \frac{3 \times 3 + 4 \times 2 - 4}{\sqrt{3^2 + 4^2}} \\ &= \frac{-3}{5} & &= \frac{13}{5} \end{aligned}$$

The signs are different.

$\therefore$ The points lie on opposite sides of the line.

Determine the shortest distance between the point and the line. Where necessary, give answers in surd form with rational denominators.

- $(2,1)$ and $3x + 4y - 5 = 0$
- $(2,1)$ and $4x - 3y + 7 = 0$
- $(3,-2)$ and $8y = 6x + 1$
- $(-1,2)$ and $5y = -12x + 3$
- $(-3,-1)$ and $4x + 3y = 9$
- $(1,4)$ and $2x + y - 4 = 0$
- $(4,-1)$ and $5x - 2y + 3 = 0$
- $(-1,-3)$ and $4 - 2x = 5y$
- $(5,-2)$ and $7x - y = -2$
- $(6,7)$ and $3y = 3x + 5$
- $(3,1)$ and $3x - 2y = 4$
- $(5,8)$ and $3x - y = -4$
- $(-5,-4)$ and $4x + 5y = 6$
- $(-3,-8)$ and $6x - 8y = 3$
- Show that the points $(1,3)$ and $(4,2)$ are on the same side of the line $x + 3y + 4 = 0$.
- Show that the points $(-1,-3)$ and $(-4,-1)$ are on opposite sides of the line $2x - 3y + 1 = 0$.
- Show that the line $3x + 4y - 20 = 0$ is a tangent to the circle $x^2 + y^2 = 16$. (Hint: Find the distance from the centre of the circle to the straight line.)
- Show that the line $3x + 4y = 15$ does not touch the circle $x^2 + y^2 = 4$.
- The points $(3,4)$ and $(-1,k)$ are the same distance from the line $3x - 2y + 6 = 0$. Determine the value(s) of k .
- Show that the line $6x - 8y = 5$ cuts the circle $(x - 3)^2 + (y - 2)^2 = 25$ in two places.

e.g. Find the equation of the line passing through the intersection of the straight lines $3x + y - 4 = 0$ and $2x - 2y + 3 = 0$ and the point $(2, -1)$.

The equation will be of the form:

$$3x + y - 4 + k(2x - 2y + 3) = 0 \quad (1) \quad \text{where } k \text{ is a constant}$$

To find k , substitute $x = 2$ and $y = -1$ into the equation.

$$\therefore 6 - 1 - 4 + k(4 + 2 + 3) = 0$$

$$\therefore 1 + 9k = 0$$

$$\therefore k = -\frac{1}{9}$$

Substituting back into (1) gives:

$$3x + y - 4 - \frac{1}{9}(2x - 2y + 3) = 0$$

$$27x + 9y - 36 - 2x + 2y - 3 = 0$$

$$\therefore 25x + 11y - 39 = 0 \text{ is the required equation.}$$

- 1 Show that the equation of the straight line passing through the intersection of the straight lines $x - y + 1 = 0$ and $2x - 3y + 1 = 0$ and the point $(1, 3)$ is $4x - 3y + 5 = 0$.
- 2 Show that the equation of the straight line passing through the intersection of the straight lines $x + y + 3 = 0$ and $x - 2y + 1 = 0$ and the point $(-1, 3)$ is $11x - 4y + 23 = 0$.
- 3 Show that the equation of the straight line passing through the intersection of the straight lines $2x + y - 4 = 0$ and $x + 3y + 5 = 0$ and the point $(1, 1)$ is $19x + 12y - 31 = 0$.
- 4 Show that the equation of the straight line passing through the intersection of the straight lines $3x + 2y - 1 = 0$ and $2x - 2y + 3 = 0$ and the point $(-2, -4)$ is $51x - 16y + 38 = 0$.
- 5 Determine the equation of the straight line passing through the intersection of the lines $2x - y + 4 = 0$ and $x + 3y - 5 = 0$ and the point $(1, 2)$ by using the equation $2x - y + 4 + k(x + 3y - 5) = 0$.
- 6 Show that the equation of the straight line passing through the intersection of the straight lines $2x + 3y + 5 = 0$ and $3x - y - 4 = 0$ and the point $(1, -3)$ is $5x + 2y + 1 = 0$.
- 7 Find the equation of the line through the intersection of the straight lines $x - y + 3 = 0$ and $2x + y - 4 = 0$ and the point $(4, 5)$.
- 8 A straight line passes through the intersection of the straight lines $x - y + 3 = 0$ and $3x + y - 4 = 0$ and the point $(-1, -4)$. Determine the equation of the line.
- 9 Find the equation of the straight line passing through the point $(1, 4)$ and the intersection of two other straight lines $3x + 2y = 5$ and $x - 4y - 3 = 0$.
- 10 A straight line passes through the intersection of the lines $x - y + 3 = 0$ and $2x + 4y - 5 = 0$ and the point $(3, 4)$. Determine the equation of the line.
- 11 A straight line passes through the intersection of the lines $x - 2y - 7 = 0$ and $2x + y = 5$ and the point $(-3, -2)$. Determine the equation of the line.
- 12 A straight line passes through the point $(5, -1)$ and the intersection of the straight lines $2x - 2y - 7 = 0$ and $3x + y - 4 = 0$. Show that the gradient of the straight line is $\frac{1}{5}$.

e.g. Find the equation of the straight line that has gradient $\frac{2}{5}$ and passes through the intersection of the straight lines $2x - y + 3 = 0$ and $3x + 2y = 7$.

The equation will be of the form: $2x - y + 3 + k(3x + 2y - 7) = 0$ ①

Rearranging terms to find the gradient: $2x - y + 3 + 3kx + 2ky - 7k = 0$

$$x(2 + 3k) + y(2k - 1) + 3 - 7k = 0$$

$$y = \frac{-2 - 3k}{2k - 1}x + \frac{7k - 3}{2k - 1}$$

$$\therefore \text{The gradient is } \frac{-2 - 3k}{2k - 1}.$$

$$\text{Equating the two gradients: } \frac{-2 - 3k}{2k - 1} = \frac{2}{5}$$

$$-10 - 15k = 4k - 2$$

$$\therefore k = -\frac{8}{19}$$

Substituting the value of k into ① $2x - y + 3 + -\frac{8}{19}(3x + 2y - 7) = 0$

$$38x - 19y + 57 - 24x - 16y + 56 = 0$$

$$\therefore 14x - 35y + 113 = 0 \text{ is the equation.}$$

- 1 Find the equation of the straight line that has a gradient of $\frac{1}{4}$ and passes through the point of intersection of the lines $x + y + 3 = 0$ and $2x - y - 1 = 0$.
- 2 Find the equation of the straight line that has a gradient of $-\frac{2}{3}$ and passes through the point of intersection of the straight lines $x - 2y + 3 = 0$ and $2x + y + 2 = 0$.
- 3 Given that a straight line is parallel to $4x + 3y - 1 = 0$ and passes through the intersection of $x - y + 4 = 0$ and $2x + 3y - 1 = 0$:
 - a Determine the gradient of the line $4x + 3y - 1 = 0$
 - b Write the gradient of the required line.
 - c Find the gradient of the straight line in terms of k .
 - d Find the value of k .
- 4 Find the equation of the line through the intersection of $x + y - 4 = 0$ and $2x - y + 1 = 0$ and parallel to the line $2x + y + 3 = 0$.
- 5 Find the equation of the line through the intersection of $2x + y + 4 = 0$ and $3x - 2y + 2 = 0$ and parallel to the line $4x - 3y - 1 = 0$.
- 6 Find the equation of the line through the intersection of $4x - y + 1 = 0$ and $3x + 2y - 1 = 0$ and parallel to the line $2y - 3x + 1 = 0$.
- 7 Find the equation of the line through the intersection of $3x + 4y - 5 = 0$ and $x - 2y - 4 = 0$ and perpendicular to the line $2x + 3y = 4$.
- 8 Find the equation of the line through the intersection of $2y + 3x + 1 = 0$ and $3x - 2y - 4 = 0$ and perpendicular to the line $x + 2y = 4$.
- 9 Find the equation of the line through the intersection of $2y - 4 = 3x$ and $3x = 4 - 2y$ and perpendicular to the line $2x - y + 4 = 0$.
- 10 Find the equation of the line through the intersection of $x - 5y + 1 = 0$ and $4x - 1 = y$ and parallel to the line joining the points $(3, 4)$ and $(-9, -1)$.
- 11 Find the equation of the line through the intersection of $2x + 4y = 5$ and $2x = 3y + 1$ and perpendicular to the line joining the points $(-4, -1)$ and $(5, -2)$.

20 concurrent lines

Concurrent lines are straight lines that all pass through the same point. To show that more than 2 straight lines are concurrent, solve two equations simultaneously to obtain a common point. Then show that this point satisfies the other equation(s).

e.g. Show that the lines with equations $x + 3y - 10 = 0$, $2x - y + 1 = 0$ and $4x - 3y + 5 = 0$ are concurrent.

$$x + 3y - 10 = 0 \quad (1)$$

$$2x - y + 1 = 0 \quad (2)$$

$$(2) \times 3$$

$$6x - 3y + 3 = 0 \quad (3)$$

$$(1) + (3)$$

$$7x - 7 = 0$$

$$\therefore x = 1$$

Substitute $x = 1$ into (1)

$$1 + 3y - 10 = 0$$

$$\therefore y = 3$$

The point of intersection is (1,3). Substitute these coordinates into the third equation:

$$4x - 3y + 5 = 0$$

$$\text{L.H.S.} = 4 \times 1 - 3 \times 3 + 5$$

$$= 0$$

$$= \text{R.H.S.}$$

This point satisfies the third equation, hence the three lines are concurrent.

Show that each set of lines is concurrent.

$$\begin{aligned} 1 \quad & 2x + 3y = 8 \\ & x - 3y = -5 \\ & 3x + y = 5 \end{aligned}$$

$$\begin{aligned} 2 \quad & x - 2y = 1 \\ & 2x + 3y = 9 \\ & 4x - y = 11 \end{aligned}$$

$$\begin{aligned} 3 \quad & 2x + 3y = 13 \\ & 3x - y = 3 \\ & 4x + y = 11 \end{aligned}$$

$$\begin{aligned} 4 \quad & x + 2y = 3 \\ & 3x - 2y = -7 \\ & 4x + 3y = 2 \end{aligned}$$

$$\begin{aligned} 5 \quad & 3x + 4y = -9 \\ & x + 2y = -5 \\ & 2x - 3y = 11 \end{aligned}$$

$$\begin{aligned} 6 \quad & 3x + 4y = -18 \\ & x + 2y = -8 \\ & x - 3y = 7 \end{aligned}$$

$$\begin{aligned} 7 \quad & 4x + 2y = 20 \\ & 3x - y = 5 \\ & x + 3y = 15 \end{aligned}$$

$$\begin{aligned} 8 \quad & 2x - y = -7 \\ & x + 3y = 7 \\ & 3x + 5y = 9 \end{aligned}$$

$$\begin{aligned} 9 \quad & 2x - y = 8 \\ & x - 3y = -6 \\ & 2x - 5y = -8 \end{aligned}$$

10 The lines $x + y = 12$, $3x - y = 4$ and $2x + 3y = c$ are concurrent. Determine the value of c .

11 The lines $x - 2y = 7$, $y - 2x = -17$ and $2x + by = 22$ are concurrent. Determine the value of b .

12 The lines $-x + y = 17$, $y + 2x = -1$ and $ay - 3x = 51$ are concurrent. Determine the value of a .

Determine whether each set of lines is concurrent.

$$\begin{aligned} 13 \quad & x + y = -1 \\ & 2x - y = -6\frac{1}{2} \\ & 3x - 2y = -10\frac{1}{2} \end{aligned}$$

$$\begin{aligned} 14 \quad & 4x + 2y = 2 \\ & 2x - 3y = 21 \\ & 3x + 4y = -11 \end{aligned}$$

$$\begin{aligned} 15 \quad & 2x + 3y = 10 \\ & x - 2y = -9 \\ & 3x + 5y = 17 \end{aligned}$$

21 linear functions: review 1

You may find the following formulae helpful.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad x = \frac{x_1 + x_2}{2}, \quad y = \frac{y_1 + y_2}{2}$$

If an equation is in the form $y = mx + b$ then m = gradient and b = y -intercept.

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{or} \quad m = \tan \theta \quad \text{where } \theta \text{ is the angle of inclination}$$

lines are parallel if $m_1 = m_2$

lines are perpendicular if $m_1 \times m_2 = -1$

Equations of a straight line

Intercept form

$$\frac{x}{a} + \frac{y}{b} = 1$$

Two-point form

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

Point/gradient form

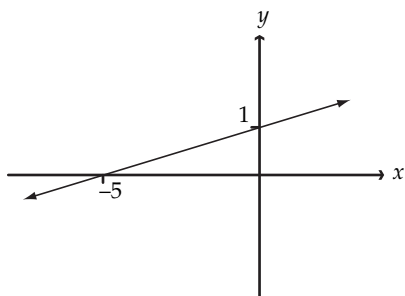
$$y - y_1 = m(x - x_1)$$

- 1 A straight line joins A(3,7) and B(5,9).
 - a Find the length of the interval AB.
 - b Determine the gradient of AB.
 - c Find the coordinates of the midpoint of AB.
 - d Determine the equation of the line AB.
- 2 Find the equation of the line that has a gradient of $\frac{2}{3}$ and passes through the point (-1,4).
- 3 Give the gradient and the x - and y -intercepts for the straight line with the equation $3x - 2y + 5 = 0$.
- 4 A straight line passes through the points (3,4) and (-1,-3). Find the size of the angle of inclination to the nearest degree.
- 5 Show that the lines $3x - 4y + 5 = 0$ and $6x = 8y - 4$ are parallel.
- 6 Give the gradient of the line that is perpendicular to the line $2x - 5y + 4 = 0$.
- 7 Find the equation of the straight line that passes through the point (4,5) and is parallel to the x -axis.
- 8 The straight line $x = 2$ passes through the point (h ,6). Give the value of h .
- 9 Draw a neat sketch of the line with the equation $2x - 3y = 12$.
- 10 Draw a neat sketch of the line with the equation $2x + 5y = 7$, labelling the x - and y -intercepts.
- 11 Does the point (3,5) lie on the line $2x = 3y - 10$?
- 12 A triangle is formed by joining the points (3,5), (-1,4) and (-2,8). Show that it is an isosceles triangle.
- 13 The points A and B have the coordinates (-1,2) and (1,4), respectively. Find the equation of the perpendicular bisector of the interval AB.
- 14 Show that the points (1,6), (-1,2) and (-3,-2) are collinear.
- 15 The lines with equations $3x - 5y + 3 = 0$ and $by - x + 4 = 0$ are perpendicular. Determine the value of b .

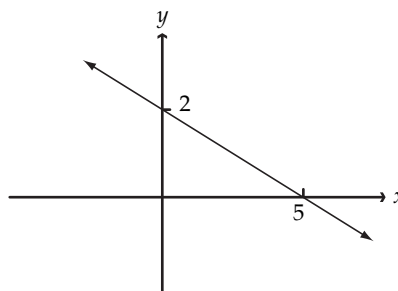
- 1 The points A(8,5), B(1,-3), C(-3,-5) and D(1,4) form a quadrilateral.
 - a Determine the length of the longer diagonal.
 - b Determine the coordinates of the midpoint of DB.
 - c Find the equation of the line AC.
- 2 A straight line has a gradient of $-\frac{3}{4}$ and passes through the point (0,-2). Find the equation of the line.
- 3 Give the gradient of a straight line that is perpendicular to the line $3x - 5y + 4 = 0$.

Give the equation of each straight line.

4

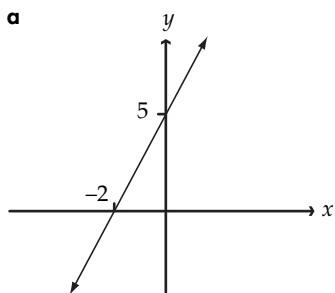


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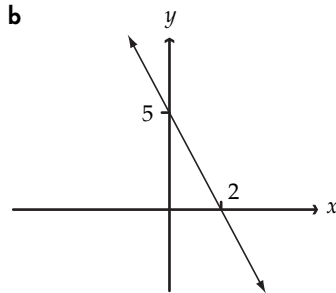


- 6 Which diagram represents the straight line $2x + 5y = 10$?

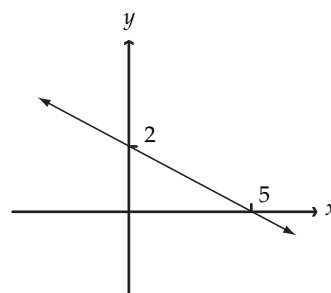
a



b



c



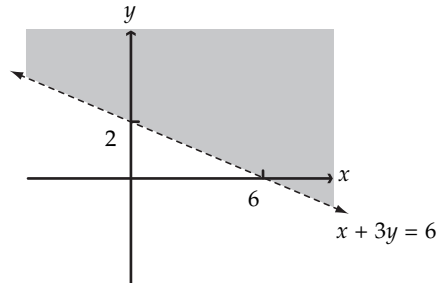
- 7 Two straight lines both pass through the point (-1,4). One line is parallel to the x-axis and the other is parallel to the y-axis. Give the equation of each line.
- 8 Which of the points A(2,3), B(-10,-3), C(3,2) and D(-2,1) lie on the straight line $2y - x = 4$?
- 9 Show that the points (1,2), (-9,-3) and (-1,1) are collinear.
- 10 Determine whether the straight line with the equation $5x - 3y = 4$ is parallel or perpendicular to the line with the equation $10y = 7 - 6x$.
- 11 Determine the equation of the straight line that passes through the point (-1,-4) and is parallel to the line $3x + 2y - 5 = 0$.
- 12 Determine the equation of the straight line passing through the intersection of the lines $2x - 5y + 7 = 0$ and $2y + 3x + 7 = 0$ and the point (4,1).
- 13 The three vertices of a triangle are A(1,3), B(2,-3) and C(-2,0).
 - a Determine the equation of BC.
 - b Determine the length of BC.
 - c Determine the distance from A to BC.
 - d Determine the area of the triangle ABC.
- 14 Determine the equation of the straight line passing through the intersection of the lines $x - 2y + 4 = 0$ and $2x + y - 3 = 0$ and perpendicular to the line $2x - 3y + 5 = 0$.
- 15 Show that the lines $x + 3y + 5 = 0$, $3x + y = 1$ and $4x - 3y = 10$ are concurrent.
- 16 A straight line makes an angle of 60° with the positive direction of the x-axis. Given that it passes through the point (1,5), write its equation.

On a Number Plane an infinite set of points lies on each side of a straight line. These sets are described by an **inequality** (or inequation).

e.g. Shade the region that satisfies the inequality $x + 3y > 6$.

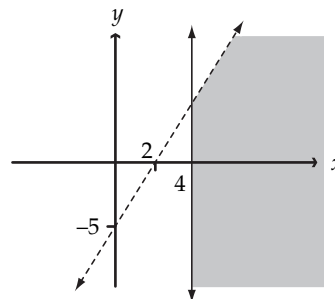
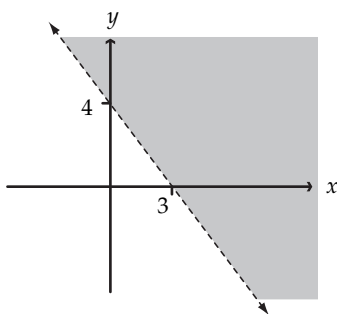
First graph the equation $x + 3y = 6$ using a broken line.

Test the point $(0,0)$ and then shade the appropriate region.



Note: With inequalities of the type $x + 3y > 9$ or $x + 2y - 5 < 0$ use a broken or dotted line. With inequalities of the type $x + 3y \geq 7$ or $x + 2y - 6 \leq 0$ use a solid line.

- 1 Show that the point $(3,4)$ satisfies the inequality $2x - 3y < 4$.
- 2 Show that the point $(-2,-1)$ satisfies the inequality $3x - 4y \leq 5$.
- 3 Show that the point $(-4,1)$ satisfies the inequality $x - 2y \leq -1$.
- 4 Which of the points $A(2,4)$, $B(-3,1)$, $C(4,-5)$ and $D(-1,-4)$ satisfy the inequality $2x - 3y > 9$?
- 5 Shade the region satisfying the inequality $x + y < 4$.
- 6 Draw a sketch to indicate the region satisfying the inequality $x - 2y > 6$.
- 7 Shade the region on the number plane defined by $x - 2y \leq -1$.
- 8 Shade the region satisfying the inequalities $x \geq 0$ and $y \geq 0$.
- 9 Shade the region defined by $2x + y < 6$, $x \geq 0$ and $y \geq 0$.
- 10 Shade the region defined by $x + y > -1$ and $x + y < 3$.
- 11 Shade the region satisfying $2x - 3y \leq 6$, $x \leq 0$ and $y \leq 0$.
- 12 Indicate on a sketch the set of points that satisfies $2x - y < 4$ and $x \geq 2$ simultaneously.
- 13 What inequality is represented by the diagram?
- 14 What inequalities define the shaded region?



- 15 Shade the region satisfying the inequalities $2x + y < 4$ and $5x - 2y \geq 10$ simultaneously.
- 16 Indicate the region defined by the inequalities $x - 3y \geq 6$ and $2x + 3y > 4$.
- 17 Shade the region satisfying $x \leq 5$, $y < 4$ and $3x + 5y > 15$ simultaneously.

Graphs of absolute equations are usually V-shaped.

e.g. Graph the function $y = |x - 1|$.

Method 1

From the definition of absolute value:

Case 1

$$\text{if } x - 1 \geq 0$$

$$\text{i.e. } x \geq 1$$

$$\text{then } y = x - 1$$

Case 2

$$\text{if } x - 1 < 0$$

$$\text{i.e. } x < 1$$

$$\text{then } y = -(x - 1)$$

$$y = -x + 1$$

Therefore in this graph:

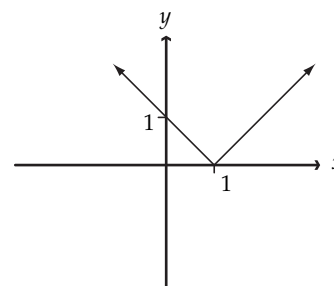
$$y = x - 1 \quad \text{if } x \geq 1$$

$$y = -x + 1 \quad \text{if } x < 1$$

Method 2

Complete the table.

x	-1	0	1	2	3
y	2	1	0	1	2



Select the correct equation for each of the four graphs from the list below.

a $y = |x + 1|$

b $y = |x - \frac{1}{2}|$

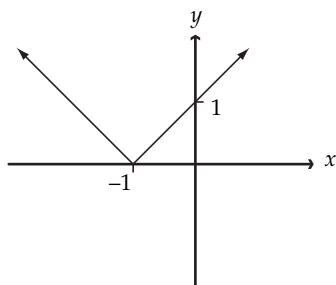
c $y = |x + \frac{1}{2}|$

d $y = |x| + 1\frac{1}{2}$

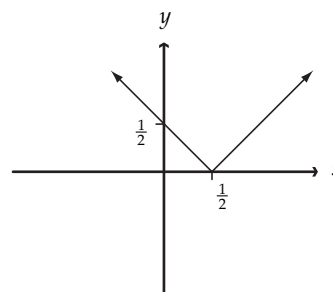
e $y = |x| - 1\frac{1}{2}$

f $y = 1\frac{1}{2} - |x|$

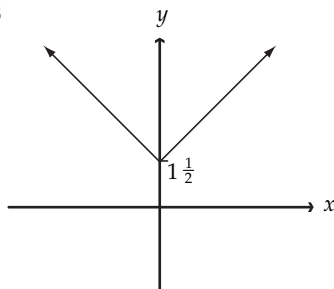
1



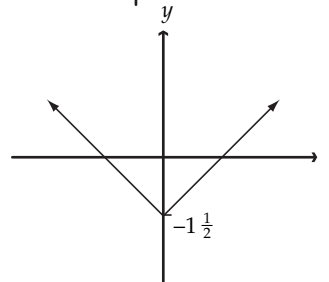
2



3



4



Draw a neat sketch of each function.

5 $y = |x| + 2$

6 $y = |x| - 2\frac{1}{2}$

7 $y = |x| + 1$

8 $y = |x| + 3$

9 $y = |x| - 1$

10 $y = |x| - 2$

11 $y = -|x|$

12 $y = 1 - |x|$

13 $y = 3 - |x|$

14 $y = |1 - x|$

15 $y = |x + 1|$

16 $y = |x + 3|$

17 $y = |x - 3|$

18 $y = |x + 2|$

19 $y = 2|x|$

20 $y = -3|x|$

21 $y = |2x + 1|$

22 $y = |2x - 2|$

23 $y = |2x| + 1$

24 $y = |2x| - 4$

Examples of function notation are $y = f(x)$, $y = 3x + 4$, $f(x) = 6x^2 - 1$, $h(x) = 7x + 3$, $H(t) = t^2 + 3t$.

e.g.

1 Given the function $f(x) = 6x^2 + 3$, evaluate $f(4)$.

$$\begin{aligned} f(4) &= 6 \times 4^2 + 3 \\ &= 99 \end{aligned}$$

Remember, for a function each value of x (or the independent variable) has only **one** value of y (the dependent variable).

Note: The graphical test for a function is to draw a vertical line through the graph. It must cut the graph at only one point.

2 Show that $x^2 + y^2 = 16$ does not represent a function.

Let $x = 3$, then:

$$3^2 + y^2 = 16$$

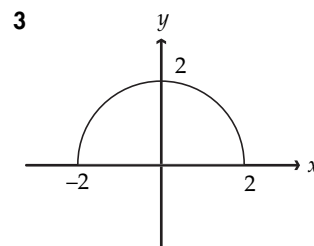
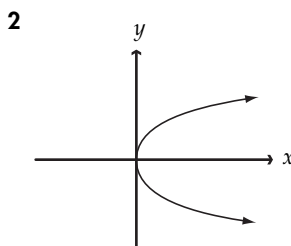
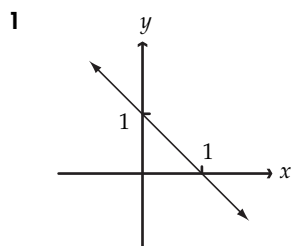
$$y^2 = 7$$

$$\therefore y = \pm\sqrt{7}$$

That is, there are 2 values for y .

$\therefore x^2 + y^2 = 16$ is not a function.

Which graphs represent functions?



- 4** Given $f(x) = 6x - 3$, find the value of $f(3)$, $f(0)$ and $f(-1)$.
- 5** Find the values of $f(-3)$, $f(2)$ and $f(6)$, given that $f(x) = 7x - 2$.
- 6** Given the function $g(x) = x^2 - x$, find the values of $g(0)$, $g(5)$ and $g(-3)$.
- 7** If $f(x) = 3x + 4$, determine $f(a)$.
- 8** Given that $H(x) = 8x^2 - 2x + 4$, determine the values of $H(2)$ and $H(-3)$.
- 9** If $f(x) = x^2 + 1$, show that $f(1) = f(-1)$.
- 10** For the function $p(x) = x^3$, show that $p(-2) = -p(2)$.
- 11** Show that $f(a) = f(-a)$, given $f(x) = 2x^2 - 3$.
- 12** Show that $f(a) = -f(-a)$, given $f(x) = 3x^3$.
- 13** If $f(x) = |6x + 1|$, determine the values of $f(-3)$ and $f(6)$.
- 14** Given that $H(x) = 6x + 3$, determine the values of x for which $H(x) = 0$ and $H(x) = 5$.
- 15** For $f(t) = \frac{t+3}{t^2+1}$, find the value of $f(3)$.
- 16** Given that $g(x) = 3^x$, find the values of $g(2)$ and $g(-1)$.
- 17** Determine the value of $f(2)$ given the function $f(x) = 3^{-x} + 3^x$.
- 18** If $f(x) = 6x + 3$ and $h(x) = 3x^2 + 4x$, determine the value of $f(3) + h(-1)$.
- 19** For $F(x) = x^4 + x^2 - 1$, show that $f(3) = f(-3)$ and $f(a) = f(-a)$.
- 20** For the function $h(t) = |t - 3| - 5$, find the values of $h(4)$ and $h(-3)$.
- 21** If $F(t) = |t^2 - 3t - 4|$, find the values of $F(8)$ and $F(3)$.

26 domain and range

When a function is defined in terms of x :

- the set of real numbers, x , for which the function is defined is called the **domain**.
- the set of values of y or $f(x)$ for all values of x in the domain is called the **range**.

That is:

- domain – set of x -values
- range – set of y -values

e.g.

1 $y = x + 1$

Domain: All reals

Range: All reals

2 $y = x^2 + 1$

Domain: All reals

Range: All reals where $y \geq 1$

3 $y = \frac{1}{x-1}$

Domain: All reals except $x = 1$

Range: All reals except $y = 0$

4 $y = \sqrt{16 - x^2}$

Domain: $-4 \leq x \leq 4$

Range: $0 \leq y \leq 4$

Give the domain of each function.

1 $f(x) = x + 3$

2 $f(x) = x^2 - 1$

3 $y = x^3$

4 $y = x^3 + x^2$

5 $y = x^4$

6 $y = x^5 + 3$

7 $f(x) = x^4 + x^3$

8 $y = x - x^4$

Give the range of the assumed domain for each function.

9 $y = x + 5$

10 $y = x^3$

11 $y = x^2$

12 $y = x^2 - 1$

13 $y = x^2 + 1$

14 $y = -x^2$

15 $y = -x^2 + 1$

16 $y = x^3 + 1$

State the domain.

17 $y = \frac{1}{x}$

18 $y = \frac{1}{x+3}$

19 $y = \frac{1}{x-3}$

20 $y = \frac{1}{1-x}$

State the range.

21 $y = \frac{1}{x}$

22 $y = \frac{1}{x-5}$

23 $y = \frac{1}{x+5}$

24 $y = \frac{1}{2-x}$

State the domain and the range.

25 $y = x^2 + 1$

26 $f(x) = \sqrt{4 - x^2}$

27 $y = -\sqrt{9 - x^2}$

28 $f(x) = \sqrt{x^2 - 4}$

29 $y = |x + 1|$

30 $y = |2x - 1|$

31 $f(x) = 3^x$

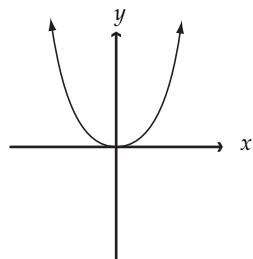
32 $y = \frac{1}{\sqrt{x+2}}$

27 odd and even functions

A function $f(x)$ is **even** if $f(a) = f(-a)$ in the given domain. A function $f(x)$ is **odd** if $f(a) = -f(-a)$ in the given domain.

e.g.

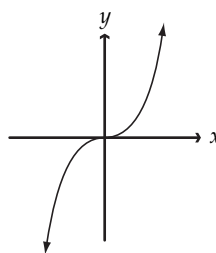
1 For $f(x) = x^2$, $f(a) = a^2$ and $f(-a) = (-a)^2$



$$= a^2$$

As $f(a) = f(-a)$, the function is even.

2 For $f(x) = x^3$, $f(a) = a^3$ and $f(-a) = (-a)^3$



$$= -a^3$$

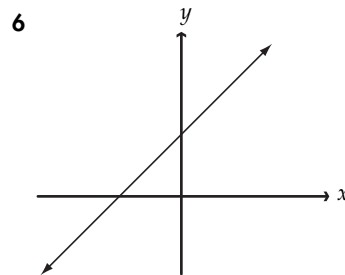
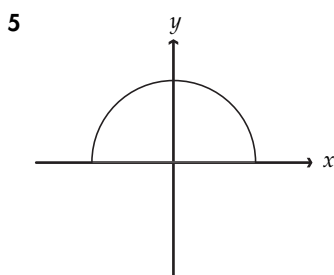
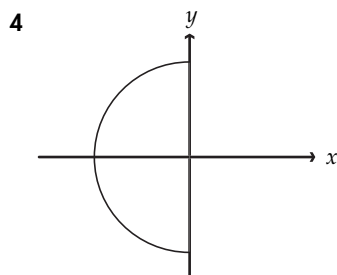
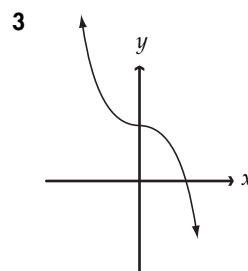
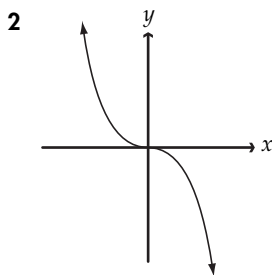
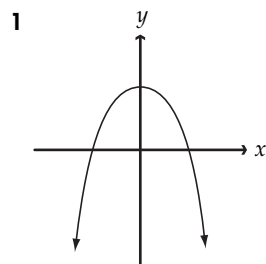
$$= -f(a)$$

As $f(a) = -f(-a)$, the function is odd.

Note: An even function is symmetrical about the y -axis. An odd function has point symmetry about the origin.

When graphing functions it helps to note whether the function is odd or even. Then it is only necessary to determine the shape of half the graph in order to sketch the complete function.

Identify whether each graph represents an odd function, an even function or neither.

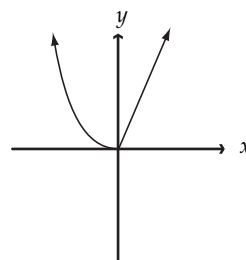


- 7** Show that $f(x) = x^2 + 3$ is even.
- 8** Show that $g(x) = x^4 + 3x^2$ is even.
- 9** Show that $f(x) = \sin x$ is odd in the domain $-180^\circ \leq x \leq 180^\circ$.
- 10** Show that $g(x) = x^5$ is an odd function.
- 11** Show that the function $f(x) = x^3 - 4$ is neither even nor odd.
- 12** Show that the function $f(x) = x^3 + x$ is odd.
- 13** Show that $g(x) = x^3 + x^2$ is neither even nor odd.
- 14** Show that $f(x) = x^2 + 4x + 1$ is neither even nor odd.
- 15** Is the function $f(x) = \cos x$ even or odd in the domain $-180^\circ \leq x \leq 180^\circ$?
- 16** Draw a sketch of $f(x) = \sin x$ in the domain $-90^\circ \leq x \leq 90^\circ$. Is the function even or odd in the stated domain?
- 17** Draw a sketch of $y = x^2 - 2$. Is the function even or odd?
- 18** Determine whether the function $y = \frac{1}{x}$ is even, odd or neither.
- 19** Show that $y = \frac{1}{x+1}$ is neither even nor odd.
- 20** Show that $y = x^4 + 3$ is even.

Some functions are made up of different functions for different sets of domains; that is, they have different expressions for different values of x .

e.g. $f(x) = 3x$ for $x > 0$
 $f(x) = x^2$ for $x \leq 0$

The graph of this function is:



$$f(1) = 3 \times 1 = 3 \quad \text{but} \quad f(-1) = (-1)^2 = 1$$

For each split function, determine the function values: $f(3)$, $f(1)$, $f(-1)$, $f(-4)$, $f(\frac{1}{2})$

1 $f(x) = 3x + 1$ for $x \geq 0$
 $f(x) = x^2 + 1$ for $x < 0$

2 $f(x) = 4x - 1$ for $x \geq 0$
 $f(x) = 1 - 3x$ for $x < 0$

3 $f(x) = 3x + 2$ for $x > 1$
 $f(x) = 9 - 4x$ for $x \leq 1$

4 $f(x) = x^2 + 1$ for $x > 0$
 $f(x) = x^2 - 1$ for $x \leq 0$

For each split function, determine the function values: $H(5)$, $H(-3)$, $H(-\frac{1}{4})$, $H(\frac{1}{4})$, $H(0)$

5 $H(x) = 3x$ for $x > 0$
 $H(x) = -x^2$ for $x \leq 0$

6 $H(x) = x^3$ for $x > 2$
 $H(x) = 8$ for $0 \leq x \leq 2$
 $H(x) = x + 8$ for $x < 0$

7 $H(x) = 6$ for $x > 2$
 $H(x) = 2x + 2$ for $0 \leq x \leq 2$
 $H(x) = x^3$ for $x < 0$

8 $H(x) = 3x + 5$ for $x > 0$
 $H(x) = 3$ for $x = 0$
 $H(x) = -2x$ for $x < 0$

For each function, determine the indicated function values.

9 $f(x) = x^2 - 5x + 4$ for $x > 4$
 $f(x) = 0$ for $0 \leq x \leq 4$
 $f(x) = x^2 - 1$ for $x < 0$
 Find $f(4)$, $f(1)$, $f(5)$, $f(-2)$

10 $f(x) = x^3$ for $x \geq 2$
 $f(x) = 3x + 2$ for $0 \leq x < 2$
 $f(x) = 2 - x$ for $x < 0$
 Find $f(5)$, $f(1)$, $f(-3)$, $f(0)$

11 $H(x) = x^2 + 5$ for $x > 2$
 $H(x) = 5$ for $-1 < x \leq 2$
 $H(x) = x - 3$ for $x \leq -1$
 Find $H(-1)$, $H(-\frac{1}{2})$, $H(\frac{1}{2})$, $H(3)$, $H(-4)$

12 $P(x) = 3x - 1$ for $x \geq 0$
 $P(x) = -1$ for $-2 \leq x < 0$
 $P(x) = 2x + 3$ for $x < -2$
 Find $P(1)$, $P(-1\frac{1}{2})$, $P(4)$, $P(-4)$

Draw a neat sketch of each function.

13 $f(x) = 2x$ for $x > 0$
 $f(x) = -x$ for $x \leq 0$

14 $f(x) = x^2 + 1$ for $x \geq 0$
 $f(x) = x + 1$ for $x < 0$

15 $H(x) = x^3 + 3$ for $x \geq 0$
 $H(x) = 3 - 2x$ for $x < 0$

16 $h(x) = 5$ for $x > 4$
 $h(x) = x + 1$ for $-1 < x \leq 4$
 $h(x) = -1$ for $x \leq -1$

To determine the point of intersection of two functions, solve the equations **simultaneously**.

e.g.

- 1** Find the point of intersection of the parabola $y = x^2 - 3x$ and the straight line $y = x - 3$.

$$y = x^2 - 3x \quad (1)$$

$$y = x - 3 \quad (2)$$

Substitute (2) into (1)

$$x - 3 = x^2 - 3x$$

$$\therefore x^2 - 4x + 3 = 0$$

$$(x - 3)(x - 1) = 0$$

$$\therefore x = 3 \text{ or } x = 1$$

If $x = 3$ then substituting into (2) gives $y = 0$

If $x = 1$ then $y = -2$

$\therefore$ The points of intersection are (3,0) and (1,-2)

- 2** Find the point(s) of intersection of the hyperbola $y = \frac{3}{x}$ and the straight line $y = x - 2$.

$$y = \frac{3}{x} \quad (1)$$

$$y = x - 2 \quad (2)$$

Substitute (2) into (1)

$$x - 2 = \frac{3}{x}$$

$$x^2 - 2x - 3 = 0$$

$$(x - 3)(x + 1) = 0$$

$$x = 3 \text{ or } -1$$

If $x = 3$ substituting into (2) gives $y = 1$

If $x = -1$, $y = -3$

$\therefore$ The points of intersection are (3,1) and (-1,-3).

Determine the point of intersection for each pair of functions.

1 $y = x^2 + 5x + 3$ and $y = 3$

3 $y = 2x - 3$ and $y = x^2 - 6x + 12$

5 $y = x + 1$ and $y = x^2 + 6x + 5$

7 $y = x - 1$ and $y = \frac{6}{x}$

9 $y = x + 4$ and $y = -\frac{3}{x}$

11 $y = x$ and $y = \frac{2}{x+1}$

13 $y = x^2 + 8x + 4$ and $y = 2x - 1$

15 $y = x - 3$ and $y = \frac{10}{x}$

2 $y = x^2 - 2x + 1$ and $y = x - 1$

4 $y = 1$ and $y = x^2 - x - 1$

6 $y = -\frac{8}{x}$ and $y = x + 6$

8 $y = \frac{10}{x}$ and $y = x - 3$

10 $y = 2x - 3$ and $y = x^2 + 8x + 5$

12 $y = \frac{1}{x}$ and $y = 3$

14 $y = x^2 - 4x + 5$ and $y = x - 1$

16 $y = x^3$ and $y = x$

30 exponential and logarithmic functions

Examples of **logarithmic** functions are: $y = \log_{10} x$, $y = \log_e x$, $f(x) = \log_4 x$

Examples of **exponential** functions are: $y = 3^x$, $y = 4^x$, $f(x) = e^x$

Note: All graphs of exponential functions of the form $y = a^x$ (where a is a constant) have the same features.

- 1 Complete the table of values for $y = \log_{10} x$ (to one decimal place).

x	0.1	0.5	1	2	3	10
$y = \log_{10} x$						

- 2 Complete the table of values for $y = \log_e x$ (to one decimal place).

x	0.1	0.5	1	2	3	10
$y = \log_e x$						

- 3 On the same set of axes, draw neat sketches of the functions $y = \log_{10} x$ and $y = \log_e x$.

- 4 On the same set of axes, draw neat sketches of $y = 2\log_{10} x$ and $y = 3\log_{10} x$.

- 5 Complete the table of values for $y = 3^x$.

x	-3	-2	-1	0	0.5	1	2
$y = 3^x$							

- 6 Complete the table of values for $y = e^x$.

x	-3	-2	-1	0	0.5	1	2
$y = e^x$							

- 7 Complete the table of values for $y = 3^{-x}$.

x	-3	-2	-1	0	0.5	1	2
$y = 3^{-x}$							

- 8 On the same set of axes, draw neat sketches of $y = 3^x$, $y = e^x$ and $y = 3^{-x}$.

- 9 On the same set of axes, draw the graphs of $y = 4^x$ and $y = 3$. Use the graphs to give the solution to the equation $3 = 4^x$.

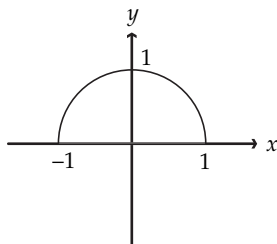
- 10 On the same set of axes, draw the graphs of $y = \log_e x$ and $y = -\frac{1}{2}$. Use the graphs to give the solution to the equation $-\frac{1}{2} = \log_e x$.

- 11 On the same set of axes, draw the graphs of $y = 4^x$, $y = \log_{10} x$ and $y = -x$. Use the graphs to give the solution to the equations $4^x = -x$ and $\log_{10} x = -x$.

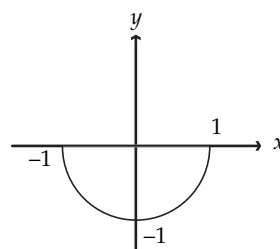
31 semi-circles and hyperbolae

The equation of a circle is not a function. However, the top and bottom semi-circles **are** functions. That is, from the equation $x^2 + y^2 = 1$, we obtain two functions: $y = +\sqrt{1-x^2}$ and $y = -\sqrt{1-x^2}$.

$$y = +\sqrt{1-x^2}$$

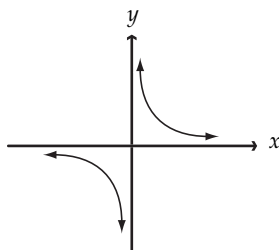


$$y = -\sqrt{1-x^2}$$

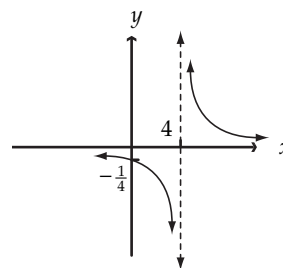


Examples of hyperbolae are $y = \frac{1}{x}$ and $y = \frac{1}{x-4}$.

$$y = \frac{1}{x}$$



$$y = \frac{1}{x-4}$$



Match one equation to each of the six graphs.

a $xy = 2$

b $y = -\frac{3}{x}$

c $y = \sqrt{x^2 - 4}$

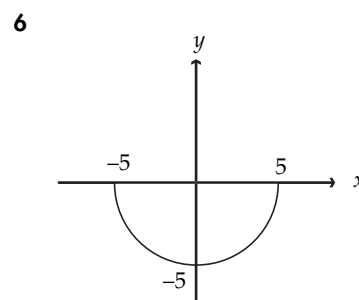
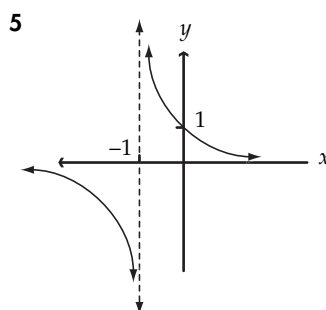
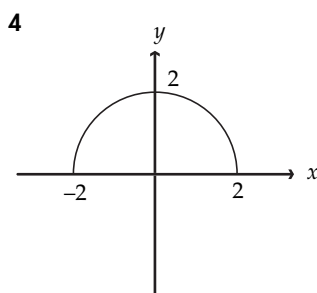
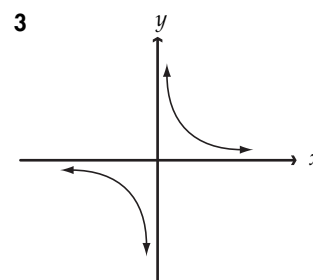
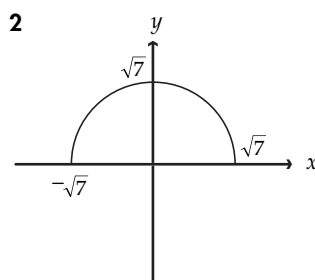
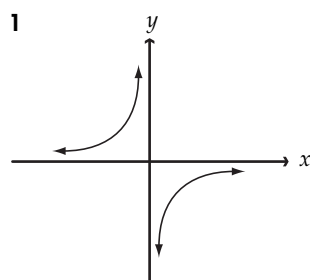
d $y = \sqrt{4 - x^2}$

e $y = -\sqrt{25 - x^2}$

f $y = \frac{1}{x+1}$

g $y = \sqrt{7 - x^2}$

h $y = \frac{1}{x-1}$



Draw a neat sketches of each equation.

7 $y = -\sqrt{16 - x^2}$

8 $xy = 4$

9 $y = \frac{1}{x-2}$

10 $y = \sqrt{9 - x^2}$

11 $y = \frac{1}{x+2}$

12 $xy = -2$

13 $y = -\sqrt{3 - x^2}$

14 $y(x-3) = 1$

15 $y = \sqrt{100 - x^2}$

16 $(x-2)y = -1$

17 $y = \frac{2}{x-3}$

18 $y = \sqrt{5 - x^2}$

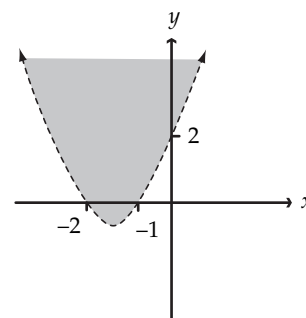
e.g.

- 1** Shade the region defined by $y > x^2 + 3x + 2$.

The boundary points are not included in the region so use the intercept method to graph the **broken** line $y = x^2 + 3x + 2$.

$$\begin{aligned} \text{Let } y &= 0 & \therefore x^2 + 3x + 2 &= 0 \\ & & (x+2)(x+1) &= 0 \\ & & x &= -1 \text{ or } x = -2 \end{aligned}$$

Test the point (0,0) in the inequality. $0 > 0^2 + 3 \times 0 + 2$ is false, so the area required is **inside** the parabola.



- 2** Shade the region satisfying the inequalities $x^2 + y^2 \leq 4$ and $y > x$.

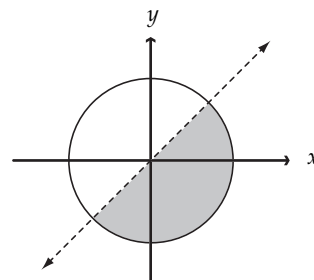
Sketch the graphs of $x^2 + y^2 = 4$ (boundary included) and $y = x$ (boundary not included).

Testing the point (0,0) in the circle: $0^2 + 0^2 \leq 4$ is true.

$\therefore$ The area required is inside the circle.

Testing the point (0,1) in the straight line gives $0 > 1$, which is false.

$\therefore$ The area required is below the line and inside the circle.



Shade the region defined by each inequality.

- | | | | |
|-----------------|-----------------------|--------------------------|---------------------------|
| 1 $y > x^2 + 1$ | 2 $y < -x^2$ | 3 $y \geq -x^2 + 2$ | 4 $x^2 + y^2 \leq 1$ |
| 5 $xy > 1$ | 6 $y > e^x$ | 7 $x^2 + y^2 \geq 25$ | 8 $y \leq \sqrt{4 - x^2}$ |
| 9 $y \leq 3^x$ | 10 $y < x^2 - 3x - 4$ | 11 $y \geq x^2 - 5x + 6$ | 12 $(x+3)^2 + y^2 \leq 1$ |

Shade the regions satisfying each pair of inequalities.

- | | |
|---------------------------------------|---|
| 13 $y > x^2 - 1$ and $y < 2$ | 14 $y \geq x^2 - 3x - 4$ and $y < -1$ |
| 15 $x^2 + y^2 \leq 36$ and $x \leq 3$ | 16 $xy > 2$ and $y > 2x$ |
| 17 $y < x^2 + 1$ and $y \geq -2$ | 18 $x^2 + y^2 \geq 4$ and $y > 1$ |
| 19 $xy < 1$, $y > 0$ and $x > 0$ | 20 $x^2 + y^2 \geq 9$ and $x^2 + y^2 \leq 36$ |

Domain – set of x -values

Range – set of y -values

A function occurs when there is only one y -value for each x -value in the domain.

A function is **even** when: $f(x) = f(-x)$

A function is **odd** when: $f(x) = -f(-x)$

1 Evaluate, given that $f(x) = 3x - 4$.

- a $f(3)$ b $f(-4)$ c $f(0)$ d $f(x+1)$

2 Evaluate, given that $g(x) = 3x^2 + 4x$.

- a $g(2)$ b $g(-3)$ c $g(x+1)$ d $g(x+h)$

3 If $f(x) = 2x + 1$ for $x \geq 0$
 $f(x) = -x + 1$ for $x < 0$ evaluate:

- a $f(-3)$ b $f(4)$ c $f(-\frac{1}{2})$ d $f(6)$

4 If $F(x) = x^3 + 1$ for $x \geq 1$
 $F(x) = 1$ for $x < 1$ evaluate:

- a $F(3)$ b $F(-1)$ c $F(\frac{1}{2})$ d $F(7)$

5 If $h(x) = x + 4$ for $x > 0$
 $h(x) = -x^2$ for $x \leq 0$

- a Evaluate $h(-3)$. b Evaluate $h(5)$. c Find the value of x if $h(x) = 10$.

Determine whether each function is odd, even or neither.

- 6 $f(x) = x^2$ 7 $g(x) = x^3$ 8 $f(x) = x^3 + 3x$ 9 $y = \sqrt{16 - x^2}$

Write the domain and range of each function.

- 10 $y = x + 1$ 11 $y = x^3$ 12 $y = \frac{1}{x+2}$ 13 $y = \frac{-1}{x+4}$

Draw a neat sketch of each function, identifying the important features.

- 14 $y = \ln x$ 15 $y = \sqrt{1 - x^2}$ 16 $y = -\sqrt{100 - x^2}$ 17 $y = \frac{1}{x-4}$

18 Determine the coordinates of the point(s) of intersection of $y = x^2 - 5$ and $y = x + 1$.

19 Shade the region defined by $xy \leq 1$.

20 Shade the region that simultaneously satisfies the inequalities $y > x^2 + 2$ and $y \leq 3$.

Domain – set of x -values

Range – set of y -values

A function occurs when there is only one y -value for each x -value in the domain.

A function is **even** when: $f(x) = f(-x)$

A function is **odd** when: $f(x) = -f(-x)$

1 Evaluate, given that $h(x) = 3x - 1$.

- a $h(3)$ b $h(-2)$ c $h(5)$ d $h(-10)$

2 Evaluate, given that $f(x) = 3 - x^2$.

- a $f(1)$ b $f(0)$ c $f(-3)$ d $f(\frac{1}{2})$

3 Evaluate, given that $H(x) = 4^x - x$.

- a $H(1)$ b $H(0)$ c $H(\frac{1}{2})$ d $H(-2)$

4 If $H(x) = 6x + 3$ and $g(x) = x^2 + 4$ evaluate:

- a $H(3) + g(2)$ b $H(4) + g(4)$ c $H(-3) + g(-2)$ d $H(-5) + g(-1)$

5 If $F(x) = 3x + 1$ for $x > 0$
 $F(x) = -x^2$ for $x \leq 0$ evaluate:

- a $F(1)$ b $F(0)$ c $F(-3)$ d $F(\frac{1}{2})$

6 If $H(x) = 2x + 1$ $x > 0$
 $H(x) = \frac{1}{2}$ $x = 0$
 $H(x) = 1 - x$ $x < 0$ evaluate:

- a $H(5)$ b $H(0)$ c $H(4)$ d $H(-5)$

Determine whether each function is odd, even or neither.

- 7 $F(x) = x^2 - 1$ 8 $h(x) = x + 2$ 9 $F(x) = x^2 + 3x + 2$ 10 $xy = 1$

Write the domain and range of each function.

- 11 $y = x^2 + 1$ 12 $y = \sqrt{16 - x^2}$ 13 $y = \frac{1}{x}$ 14 $y = |x + 3|$

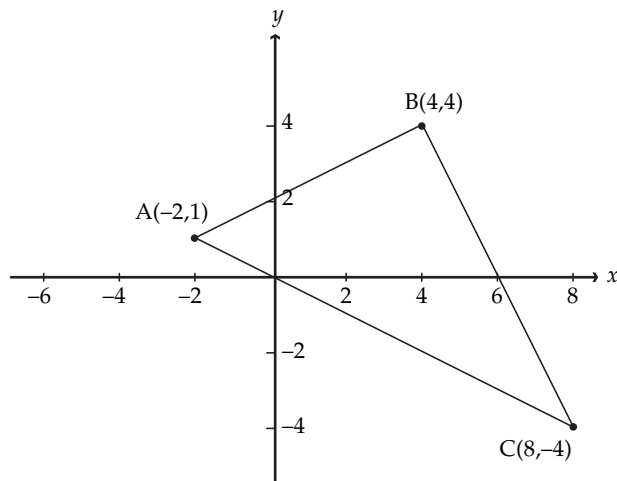
Draw a neat sketch of each function, identifying the important features.

- 15 $y = e^x$ 16 $y = x^2 + 5x + 6$ 17 $y = x^3 + 1$ 18 $y = \frac{1}{x+1}$

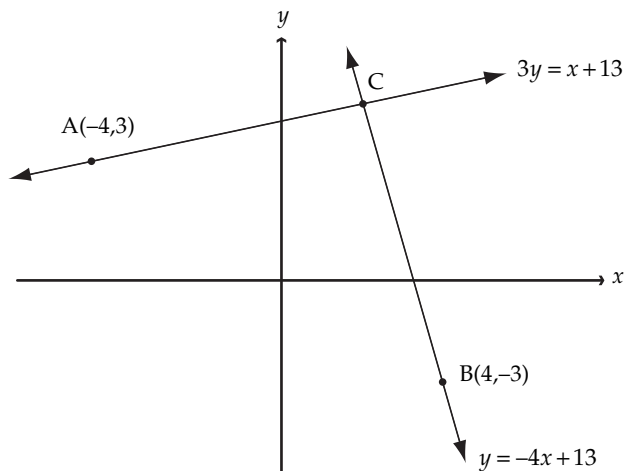
19 Determine the coordinates of the point of intersection of $y = \frac{2}{x+1}$ and $y = x$.

20 Shade the region defined by $y \geq x^2 - 2$.

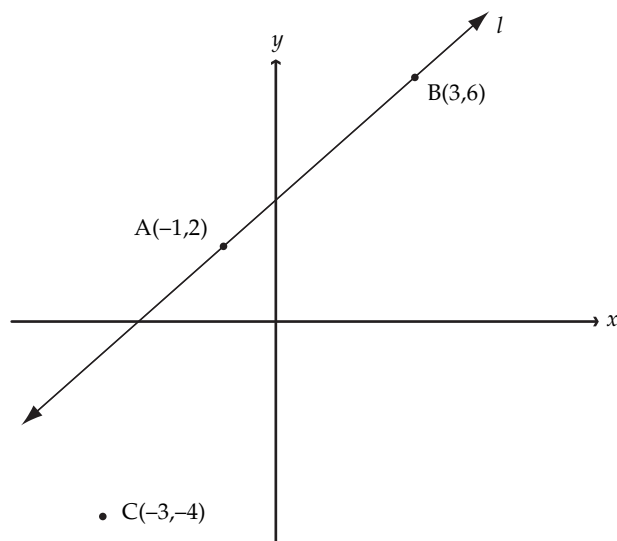
21 Shade the region that simultaneously satisfies the inequalities $y \geq e^x$, $x > -1$ and $y \leq 3$.



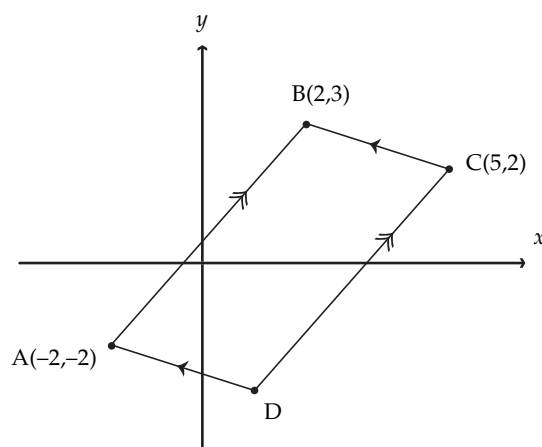
- 1 The points $A(-2,1)$, $B(4,4)$ and $C(8,-4)$ form a triangle.
- Show that the equation of the line AB is $y = \frac{1}{2}x + 2$.
 - Show that AB is perpendicular to BC .
 - If D is the midpoint of AB and E is the midpoint of AC , determine the coordinates of D and E .
 - Show that DE is parallel to BC .
 - Show that $2 \times DE = BC$.
 - Determine the acute angle (to the nearest degree) which the line AB makes with the x -axis.



- 2 The line $3y = x + 13$ passes through the points $A(-4,3)$ and C . The line $y = -4x + 13$ passes through the points $B(4,-3)$ and C .
- Determine the coordinates of C .
 - Determine θ , the obtuse angle between line BC and the x -axis.
 - If D is the midpoint of AB and E is the midpoint of BC , show that DE is parallel to AC .
 - Find the perpendicular distance from B to the line AC .
 - Calculate the length of AC , leaving the answer as a surd.
 - Show that the area of $\triangle ABC$ is 26 square units.

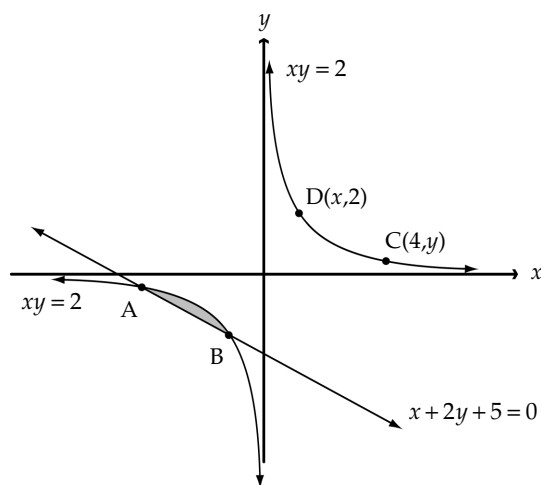


- 1 The points $A(-1, 2)$ and $B(3, 6)$ both lie on the line l . The point C has coordinates $(-3, -4)$.
 - a Determine the equation of the line l .
 - b Find the length of the interval AB as a surd.
 - c Determine the equation of the line m , passing through C and parallel to the line l . (Answer in surd form.)
 - d The point $D(x, 4)$ lies on the line m . Find the value of x .
 - e Calculate the perpendicular distance (as a surd) from the point C to the line l .
 - f Calculate the area of $\triangle ABC$.



- 2 The points $A(-2, -2)$, $B(2, 3)$, $C(5, 2)$ and D form a parallelogram.
 - a Give the coordinates of D .
 - b Determine the equation of l , the line joining A to C .
 - c Show that the line l passes through the midpoint M of the interval BD .
 - d The point E lies on the line l such that DE is perpendicular to AC . Determine the length of DE .
 - e Calculate the area of $\triangle ACD$ and hence give the area of the parallelogram $ABCD$.

- 1 The points $A(-1, y)$, $B(2, -1)$, $C(1, -7)$ and $D(-5, 1)$ form a quadrilateral.
- Show that the equation of CD is $4x + 3y + 17 = 0$.
 - Given that AB is parallel to CD , determine the value of y .
 - Calculate the length of CD .
 - Determine the perpendicular distance from B to DC .
 - Given that the length of AB is 5 units, calculate the area of $ABCD$.
 - Give the acute angle, to the nearest degree, between the y -axis and the line AB .



- 2 The hyperbola $xy = 2$ and the straight line $x + 2y + 5 = 0$ intersect at A and B .
- Determine the coordinates of A and B .
 - Given that $x < 0$, give the inequalities represented by the shaded area.
 - The points $C(4, y)$ and $D(x, 2)$ lie on the hyperbola. Give the values of x and y .
 - Determine the length of AB , leaving the answer as a surd.
 - Show that AB is parallel to DC .
 - Given that $AD \parallel BC$, what type of quadrilateral is formed by joining the points A , B , C and D ?
 - Determine, as a surd, the distance from D to AB .
 - Calculate the area of the quadrilateral $ABCD$.

1 distance formula 1 5 2 10 3 $5\sqrt{2}$ 4 $2\sqrt{17}$

5 $\sqrt{26}$ 6 5 7 $4\sqrt{5}$ 8 $\sqrt{157}$ 9 $2\sqrt{2}$ 10 $5\sqrt{2}$ 11 $\sqrt{85}$
 12 $\sqrt{34}$ 13 $2\sqrt{10}$ 14 $\sqrt{41}$ 15 5.4 16 14.2 17 11.0 18 5.1
 19 8.9 20 3.6 21 9.2 22 11.7 28 -3 or 21

2 midpoint formula 1 (6,4) 2 (5,5) 3 (4,5) 4 $(5\frac{1}{2}, 1\frac{1}{2})$

5 (2,2) 6 $(1\frac{1}{2}, -3)$ 7 $(3\frac{1}{2}, 4)$ 8 $(-3, 6)$ 9 $(3, -4\frac{1}{2})$ 10 $(-7\frac{1}{2}, 2)$
 11 $(1\frac{1}{2}, 2\frac{1}{2})$ 12 $(\frac{1}{2}, \frac{1}{2})$ 13 $(\frac{1}{2}, -5)$ 14 (1,1) 15 $(-4\frac{1}{2}, -7\frac{1}{2})$
 16 (5, -17) 17 $(-9\frac{1}{2}, -7\frac{1}{2})$ 18 $(-10\frac{1}{2}, -6)$ 19 $x=8, y=10$
 20 $x=8, y=-2$ 21 $x=10, y=8$ 22 $(-6, -1)$ 23 (2,5) 24 $(5, 9\frac{1}{2})$
 25 $(9\frac{1}{2}, 10\frac{1}{2})$ 26 (1, -3) 27 (16, -4)

3 gradient formula 1 + 2 + 3 - 4 + 5 + 6 - 7 1

8 $-\frac{1}{7}$ 9 1 10 $-\frac{13}{2}$ 11 $\frac{14}{11}$ 12 -5 13 $-\frac{5}{3}$ 14 3 15 $y=3$ 16 $x=6$
 17 $x=4$ 21 No

4 gradient: parallel and perpendicular lines,

angles of inclination 1 $-\frac{6}{5}$ 2 -2 7 parallel 8 $\frac{1}{\sqrt{3}}$

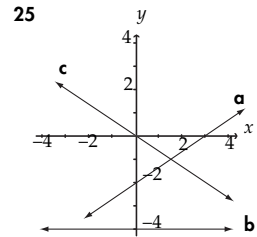
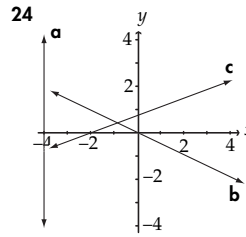
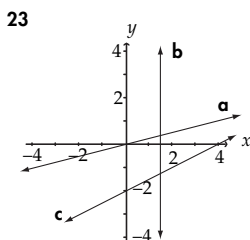
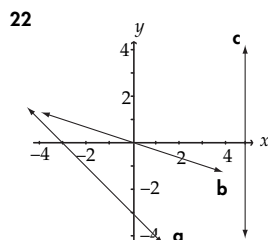
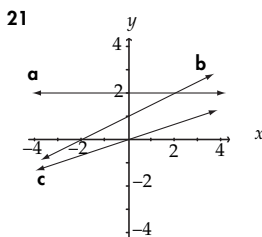
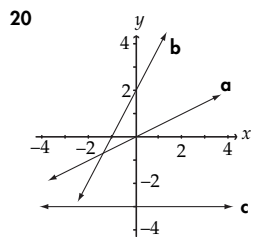
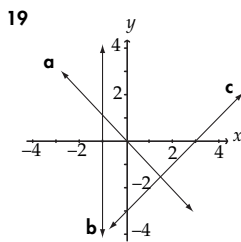
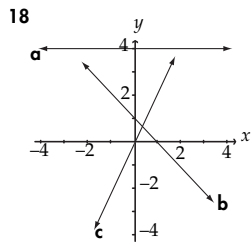
9 $\sqrt{3}$ 10 $-\frac{1}{\sqrt{3}}$ 11 -1 12 2.7 13 0.8 14 -2.1 15 -0.4 16 27°
 17 58° 18 128° 19 100° 20 1 21 $-\sqrt{3}$ 22 0.4 23 -0.5
 24 -0.7 25 1.2 26 45° 27 50° 28 135° 29 $y=5$ 30 $x=10$

5 distance, midpoint and gradient review

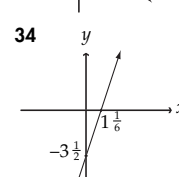
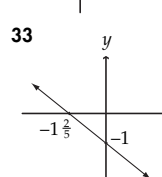
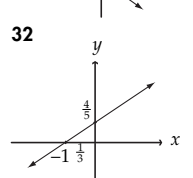
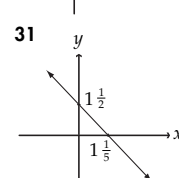
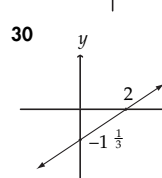
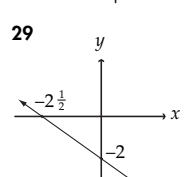
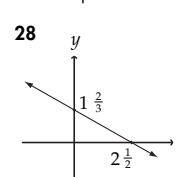
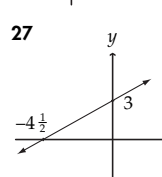
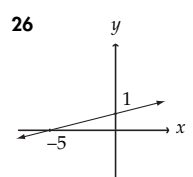
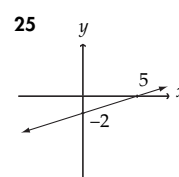
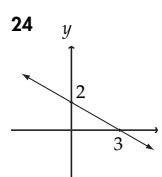
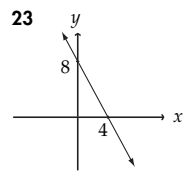
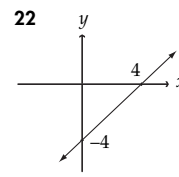
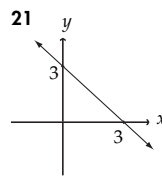
1 a $\sqrt{41}$ b $(5, 7\frac{1}{2})$ c $\frac{5}{4}$ d 51° 2 a $\sqrt{13}$ b $(-4, 2\frac{1}{2})$ c $-\frac{3}{2}$
 d 124° 3 $y=5$ 6 S(3,5) 7 D(4,4) 8 29° 9 $x=3$

6 introduction to straight lines 1 parallel to x -axis

2 through the origin 3 parallel to y -axis 4 parallel to y -axis
 5 cuts both axes 6 cuts both axes 7 cuts both axes 8 cuts both
 axes 9 parallel to y -axis 10 parallel to x -axis 11 through the
 origin 12 cuts both axes 13 cuts both axes 14 cuts both axes
 15 through the origin 16 through the origin 17 $y=0, x=0$

**7 using intercepts to graph straight lines** 1 4, 4

2 -6, -6 3 3, 6 4 1, $1\frac{1}{2}$ 5 3, $-4\frac{1}{2}$ 6 $2\frac{1}{3}$, -7 7 5, $-2\frac{1}{2}$
 8 $3\frac{1}{2}$, $2\frac{1}{3}$ 9 2, $\frac{4}{5}$ 10 -4, 4 11 -2, 6 12 4, 6 13 -3, $1\frac{1}{2}$
 14 $1\frac{1}{4}$, $-2\frac{1}{2}$ 15 $\frac{1}{3}$, $-\frac{1}{2}$ 16 1, $\frac{3}{2}$ 17 $\frac{4}{5}$, -2 18 $\frac{2}{5}$, $\frac{2}{7}$
 19 $-\frac{5}{4}$, $\frac{5}{3}$ 20 $-\frac{3}{2}$, $\frac{6}{5}$

**8 general equations: finding the gradient and y-intercept** 1 -2, -3 2 -3, 4 3 2, 5 4 -1, $\frac{3}{2}$

5 $-\frac{2}{3}$, $-\frac{5}{3}$ 6 $\frac{2}{3}$, $-\frac{4}{3}$ 7 $-\frac{4}{3}$, $\frac{2}{3}$ 8 -3, $\frac{5}{2}$ 9 $\frac{3}{2}$, $-\frac{5}{2}$
 10 $-\frac{7}{4}$, $\frac{3}{4}$ 11 $-\frac{3}{4}$, 0 12 $\frac{2}{3}$, 0 13 $3x-y+4=0$ 14 $2x-y+1=0$
 15 $5x-y+2=0$ 16 $3x+y+1=0$ 17 $2x-3y+12=0$ 18 $x-2y+6=0$
 19 $3x+4y-20=0$ 20 $2x+5y+15=0$ 21 $2x+3y+2=0$
 22 $3x-6y+2=0$ 23 $2x-y+4=0$ 24 $2x-3y-6=0$
 25 $5x-4y-20=0$ 26 $4x+3y-12=0$ 27 $5x+2y+10=0$
 28 $x+6y-6=0$ 29 $x-y+2=0$ 30 $x+2y-3=0$ 31 $x+6y+3=0$
 32 $2x-6y-3=0$ 33 $6x-2y-3=0$ 34 $x+4y-6=0$

9 point-gradient formula for the equation of a straight line 1 $3x-y-8=0$ 2 $4x-y-13=0$ 3 $x+y=0$

4 $4x+y+1=0$ 5 $3x-4y-7=0$ 6 $2x-3y-6=0$ 7 $x+3y+6=0$
 8 $3x-2y-6=0$ 9 $5x+2y+38=0$ 10 $2x-y+4=0$
 11 $3x-2y+13=0$ 12 $2x-y-5=0$ 13 $x-2y-7=0$ 14 $3x-4y-29=0$
 15 $3x+y+5=0$ 16 $2x+3y+11=0$ 17 $7x+3y+12=0$
 18 $x-y+1=0$ 19 $\sqrt{3}x-y+5+2\sqrt{3}=0$ 20 $\sqrt{3}x+y-2=0$

$$21x + \sqrt{3}y + 4 - \sqrt{3} = 0$$

10 two-point formula for the equation of a straight line

1 $2x - y + 2 = 0$ 2 $4x - 3y + 10 = 0$
 3 $4x - 3y + 16 = 0$ 4 $5x - 4y - 7 = 0$ 5 $7x + 2y - 22 = 0$ 6 $3x - 2y + 5 = 0$
 7 $x - 2y - 7 = 0$ 8 $2x + 3y + 16 = 0$ 9 $x + 6y - 24 = 0$ 10 $8x - 5y - 24 = 0$
 11 $5x - y - 19 = 0$ 12 $x - y + 1 = 0$ 13 $4x - 3y = 0$ 14 $x - y + 2 = 0$
 15 $4x - 9y - 32 = 0$ 16 $x - 4y + 39 = 0$ 17 $3x + 2y - 26 = 0$
 18 $3x + 2y - 1 = 0$ 19 $y = \frac{2}{7}x - \frac{13}{7}$ 20 $y = -\frac{11}{4}x + \frac{5}{4}$ 21 $y = \frac{3}{5}x + \frac{11}{5}$,
 $(0, \frac{11}{5})$ 22 $y = 2x$, y -intercept is 0, i.e. there is no constant

11 intercept formula and lines parallel to the axes

1 $3x - 4y + 12 = 0$ 2 $x + 2y - 4 = 0$ 3 $5x + 3y + 15 = 0$
 4 $x - 2y + 2 = 0$ 5 $x = 4$ 6 $y = 3$ 7 $x = 2$ 8 $y = -4$ 9 $x = -2$
 10 $x - 3y - 3 = 0$ 11 $5x - 3y + 15 = 0$ 12 $3x + 2y + 6 = 0$
 13 $4x + 3y - 6 = 0$ 14 $y = -3$ 15 $y = 4$ 16 $x = -3$ 17 $x = -4$
 18 $x = -3$ 19 $y = 4$ 20 $x = 4$ 21 $2x + y - 4 = 0$ 22 $2x + y - 5 = 0$
 23 $y = \pm 3$ 24 $x = 5$ 25 $y = -2$ 26 $x = 3$, $y = -1$ 27 $3x + 4y - 12 = 0$,
 $3x - 4y - 12 = 0$ 28 $5x - 4y + 20 = 0$ 29 $x - 3y - 3 = 0$
 30 $4x + 7y - 28 = 0$ 31 $y = 4$ 32 $x = -2$ 33 $x = 3$

12 finding equations of straight lines: review

1 $3x - 2y + 5 = 0$ 2 $11x - 4y - 1 = 0$ 3 $3x + 5y - 13 = 0$
 4 $x + 6y - 41 = 0$ 5 $x - 2y + 8 = 0$ 6 $3x + 2y - 6 = 0$ 7 $7x + 4y + 8 = 0$
 8 $5x - y - 10 = 0$ 9 $4x + y - 10 = 0$ 10 $2x - 5y + 7 = 0$ 11 $x - 2y + 7 = 0$
 12 $5x - 6y - 1 = 0$ 13 $3x + 2y - 5 = 0$ 14 $x - 4y - 4 = 0$
 15 $4x - 3y + 12 = 0$ 16 $5x - 2y + 10 = 0$ 17 $2x + 3y - 9 = 0$
 19 $5x - 3y = 0$ 20 $y = 5$ 21 $x = 3\frac{1}{2}$ 22 $x = 7$ 23 $y = 7$ 24 $x = -3$
 25 $y = 4$, $6x - 7y + 4 = 0$

13 determining whether a point lies on a line

5 yes 7 yes 9 $a = 2\frac{1}{3}$ 10 $x_1 = -7$ 11 $c = -11$ 13 $(-1, -3)$
 14 $x - y + 1 = 0$ 18 $h = -\frac{1}{2}$

14 intersection of two straight lines - simultaneous equations

1 $(1, 2)$ 2 $(-1, 3)$ 3 $(-1, 2)$
 4 $(4, -1)$ 5 $(-2, 3)$ 6 $(4, 1)$ 7 $(6\frac{1}{2}, 1\frac{1}{2})$ 8 $(3, 2)$ 9 $(\frac{1}{2}, 3)$
 10 $(1, 3)$ 11 $(-1, 3)$ 12 $(1, 4)$ 13 $(-1, -2)$ 14 $(2\frac{1}{2}, 0)$, $(2, 1)$,
 $(-1, 0)$ 15 $(0, 0)$, $(-1, 2)$, $(2, 4)$ 16 $(1, -3)$, 17 $(-3, 1)$

15 parallel and perpendicular lines

1 parallel 2 neither 3 perpendicular 4 neither 5 neither
 6 perpendicular 7 parallel 8 parallel 9 perpendicular
 10 neither 11 parallel 12 parallel 15 $h = 2$ 16 $k = 9$

16 collinear points - using the gradient

19 $h = 1$ 20 $k = 6$ 21 $k = -4$ 22 $h = -1\frac{1}{2}$ 23 $k = -4\frac{1}{3}$

17 perpendicular distance from a point to a

straight line 1 1 2 $2\frac{2}{5}$ 3 $\frac{\sqrt{35}}{10}$ 4 $\frac{5}{13}$ 5 $4\frac{4}{5}$

6 $\frac{2\sqrt{5}}{5}$ 7 $\frac{25\sqrt{29}}{29}$ 8 $\frac{21\sqrt{29}}{29}$ 9 $\frac{39\sqrt{2}}{10}$ 10 $\frac{\sqrt{2}}{3}$ 11 $\frac{3\sqrt{13}}{13}$ 12 $\frac{11\sqrt{10}}{10}$
 13 $\frac{46\sqrt{41}}{41}$ 14 $4\frac{3}{10}$ 19 $k = -2, 5$

18 equation of a line through the intersection of two straight lines

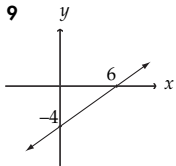
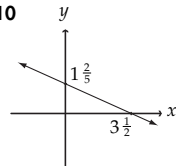
1 $y = 2$ 7 $5x - 11y + 35 = 0$
 8 $29x - 5y + 9 = 0$ 9 $5x + y - 9 = 0$ 10 $13x - 25y + 61 = 0$
 11 $x - 32y - 61 = 0$

19 equation of a line through the intersection of two straight lines

1 $3x - 12y - 26 = 0$
 2 $10x + 15y + 2 = 0$ 3 $a = \frac{4}{3}$ $b = \frac{4}{3}$ $c = \frac{-(1+2k)}{3k-1}$ $d = \frac{7}{6}$
 4 $2x + y - 5 = 0$ 5 $28x - 21y + 16 = 0$ 6 $33x - 22y + 17 = 0$
 7 $15x - 10y - 46 = 0$ 8 $8x - 4y - 9 = 0$ 9 $x + 2y - 4 = 0$ 10
 $95x - 228y + 30 = 0$ 11 $126x - 14y - 163 = 0$

20 concurrent lines 10 $c = 32$ 11 $b = 4$ 12 $a = 3$ 13 yes
 14 yes 15 yes

21 linear functions: review

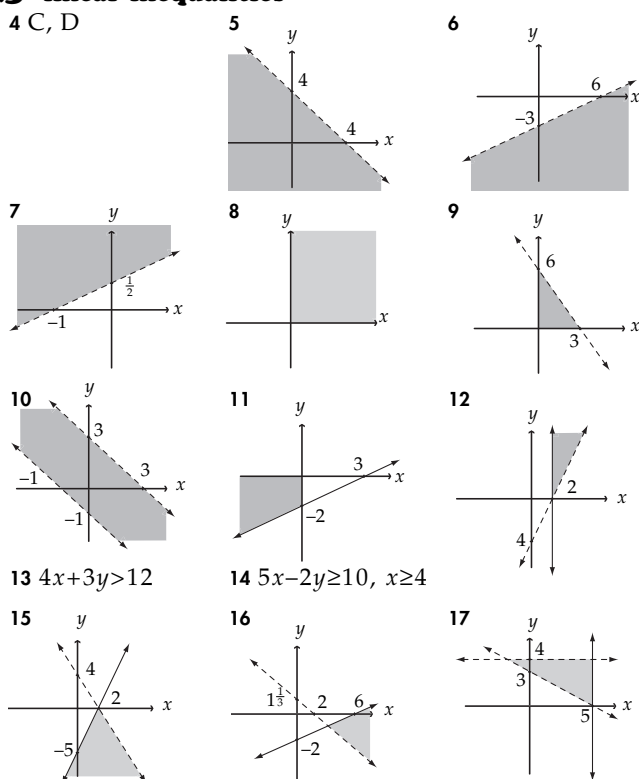
1 a $2\sqrt{2}$ b 1 c $(4, 8)$
 d $x - y + 4 = 0$ 2 $2x - 3y + 14 = 0$ 3 $\frac{3}{2}, -\frac{5}{3}, \frac{5}{2}$ 4 60° 6 $-\frac{5}{2}$
 7 $y = 5$ 8 $h = 2$ 9  10 
 11 No 13 $x - y + 3 = 0$
 15 $b = -\frac{3}{5}$

22 linear functions: review

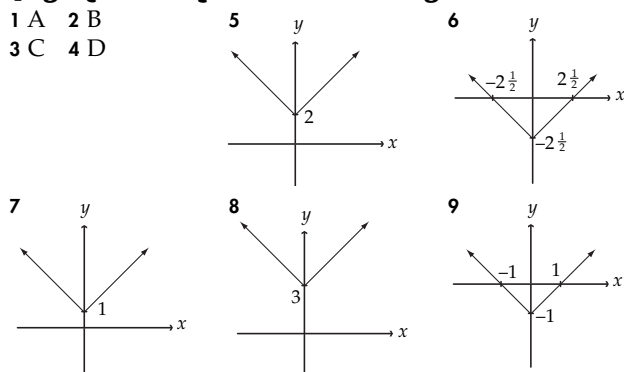
1 a $AC = \sqrt{221}$ b $(1, \frac{1}{2})$
 c $10x - 11y - 25 = 0$ 2 $3x + 4y + 8 = 0$ 3 $m = -\frac{5}{3}$ 4 $x - 5y + 5 = 0$
 5 $2x + 5y - 10 = 0$ 6 C 7 $y = 4$, $x = -1$ 8 A, B, D 10 perpendicular
 11 $3x + 2y + 11 = 0$ 12 $12x - 125y + 77 = 0$ 13 a $3x + 4y + 6 = 0$ b 5
 c $\frac{21}{5}$ d $11\frac{1}{2}$ square units 14 $15x + 10y - 28 = 0$
 16 $\sqrt{3}x - y - \sqrt{3} + 5 = 0$

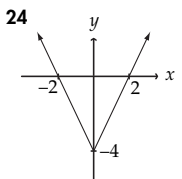
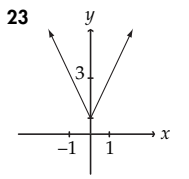
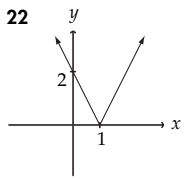
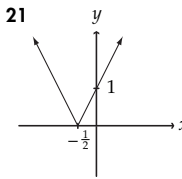
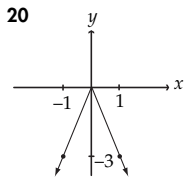
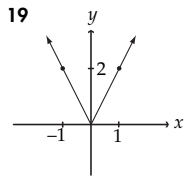
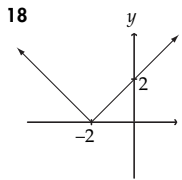
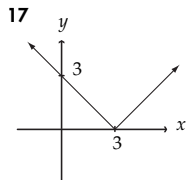
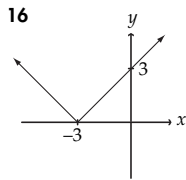
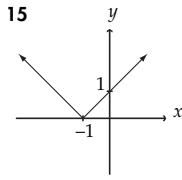
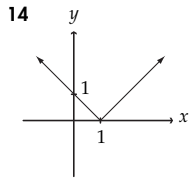
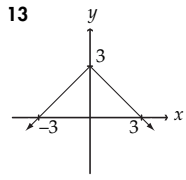
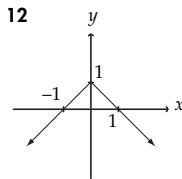
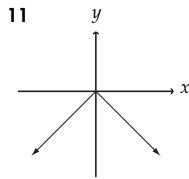
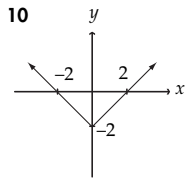
23 linear inequalities

4 C, D



24 graphs of equations involving absolute values

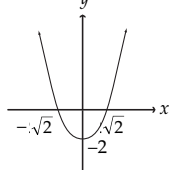
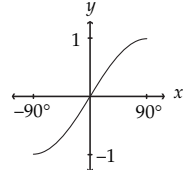




25 function notation 1 yes 2 no 3 yes 4 15, -3, -9
5 -23, 12, 40 6 0, 20, 12 7 $3a+4$ 8 32, 82 13 17, 37
14 $-\frac{1}{2}, \frac{1}{3}$ 15 $\frac{3}{5}$ 16 9, $\frac{1}{3}$ 17 $9\frac{1}{9}$ 18 20 20 -4, 1 21 36, 4

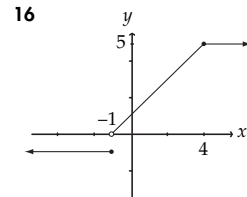
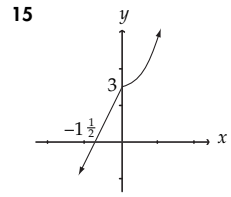
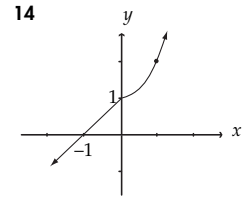
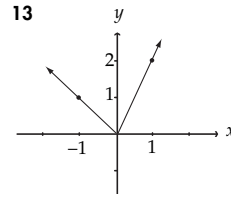
26 domain and range 1 all reals 2 all reals 3 all reals
4 all reals 5 all reals 6 all reals 7 all reals 8 all reals
9 all reals 10 all reals 11 all reals where $y \geq 0$ 12 all reals
where $y \geq -1$ 13 all reals where $y \geq 1$ 14 all reals where $y \leq 0$
15 all reals where $y \leq 1$ 16 all reals 17 all reals except $x=0$
18 all reals except $x=-3$ 19 all reals except $x=3$ 20 all reals
except $x=1$ 21 all reals except $y=0$ 22 all reals except $y=0$
23 all reals except $y=0$ 24 all reals except $y=0$ 25 all reals,
all reals where $y \geq 1$ 26 all reals where $-2 \leq x \leq 2$, all reals
where $0 \leq y \leq 2$ 27 all reals where $-3 \leq x \leq 3$, all reals where
 $-3 \leq y \leq 0$ 28 all reals where $x \leq -2$, $x \geq 2$, all reals where $y \geq 0$
29 all reals, all reals where $y \geq 0$ 30 all reals, all reals where
 $y \geq 0$ 31 all reals, all reals where $y > 0$ 32 all reals where
 $x > -2$, all reals where $y > 0$

27 odd and even functions 1 even 2 odd 3 neither
4 neither 5 even 6 neither 15 even
16 odd 17 even 18 odd



28 split functions 1 10, 4, 2, 17, $2\frac{1}{2}$ 2 11, 3, 4, 13,
 $-\frac{1}{2}$ 3 11, 5, 13, 25, 7 4 10, 2, 0, 15, $1\frac{1}{4}$ 5 15, -9, $-\frac{1}{16}$,
 $\frac{3}{4}$, 0 6 125, 5, $7\frac{3}{4}$, 8, 8 7 6, -27, $-\frac{1}{64}$, $2\frac{1}{2}$, 2 8 20, 6,
 $-\frac{1}{2}$, $5\frac{3}{4}$, 3 9 0, 0, 4, 3 10 125, 5, 5, 2 11 -4, 5, 5, 14, -7

12 2, -1, 11, -5



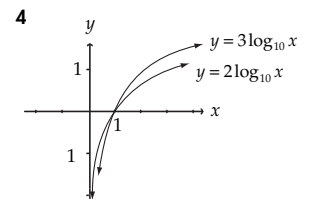
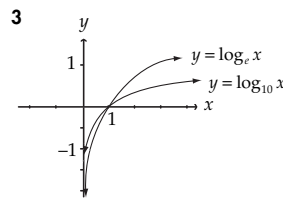
29 intersection of linear and non-linear graphs

1 (0,3) (-5,0) 2 (1,0) (2,1) 3 (5,7) (3,3) 4 (2,1) (-1,1) 5 (-1,0)
(-4,-3) 6 (-2,4) (-4,2) 7 (3,2) (-2,-3) 8 (-2,-5) (5,2) 9 (-1,3)
(-3,1) 10 (-2,-7) (-4,-11) 11 (-2,-2) (1,1) 12 $(\frac{1}{3}, 3)$ 13 (-1,-3)
(-5,-11) 14 (3,2) (2,1) 15 (5,2) (-2,-5) 16 (0,0) (1,1) (-1,-1)

30 exponential and logarithmic functions

1	x	0.1	0.5	1	2	3	10
	$y = \log_{10} x$	-1	-0.3	0	0.3	0.5	1

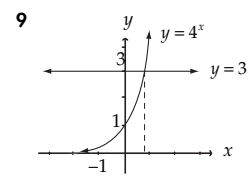
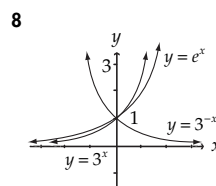
2	x	0.1	0.5	1	2	3	10
	$y = \log_e x$	-2.3	-0.7	0	0.7	1.1	2.3



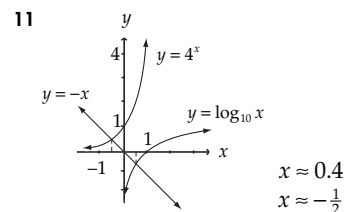
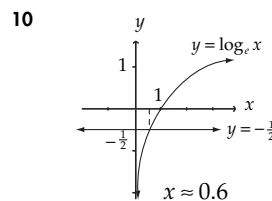
5	x	-3	-2	-1	0	0.5	1	2
	$y = 3^x$	0.04	0.1	0.3	0	1.7	3	9

6	x	-3	-2	-1	0	0.5	1	2
	$y = e^x$	0.05	0.1	0.4	1	1.6	2.7	7.4

7	x	-3	-2	-1	0	0.5	1	2
	$y = 3^{-x}$	27	9	3	1	0.6	0.3	0.1

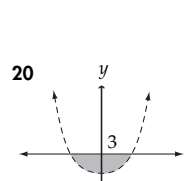
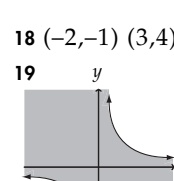
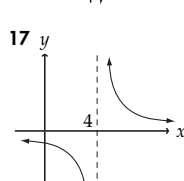
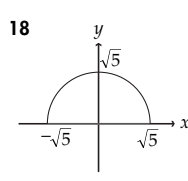
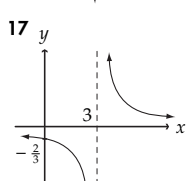
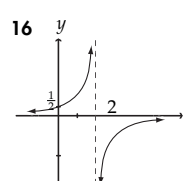
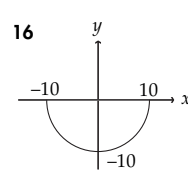
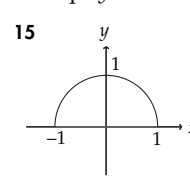
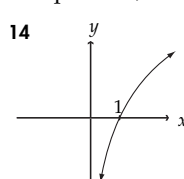
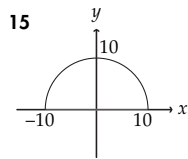
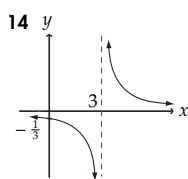
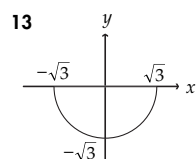
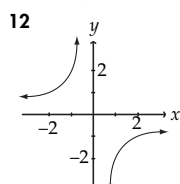
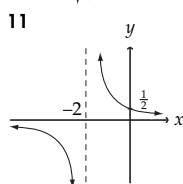
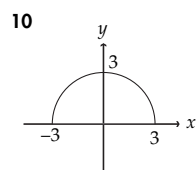
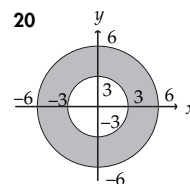
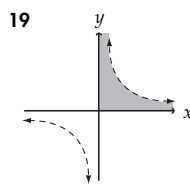
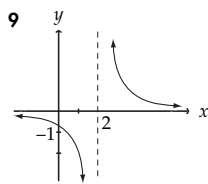
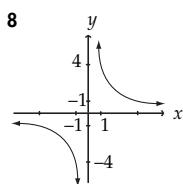
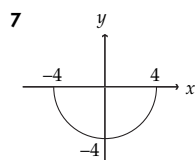


$x \approx 0.8$

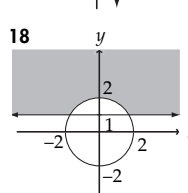
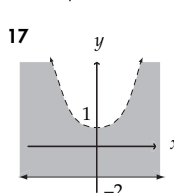
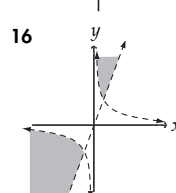
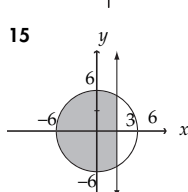
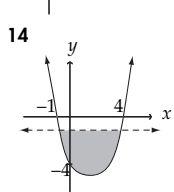
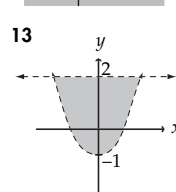
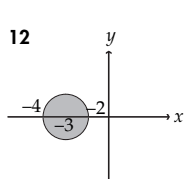
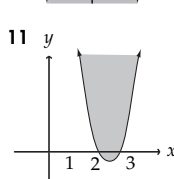
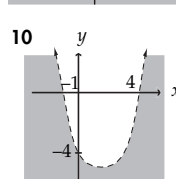
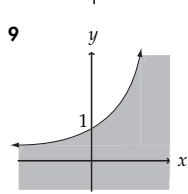
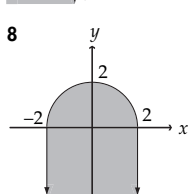
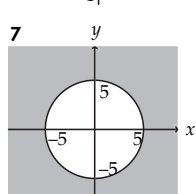
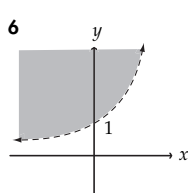
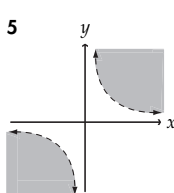
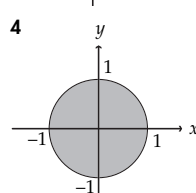
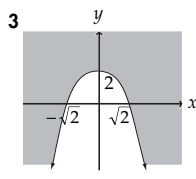
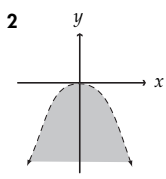
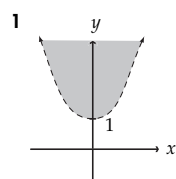


$x \approx 0.4$
 $x \approx -\frac{1}{2}$

31 semi-circles and hyperbolae 1 B 2 G 3 A 4 D
5 F 6 E

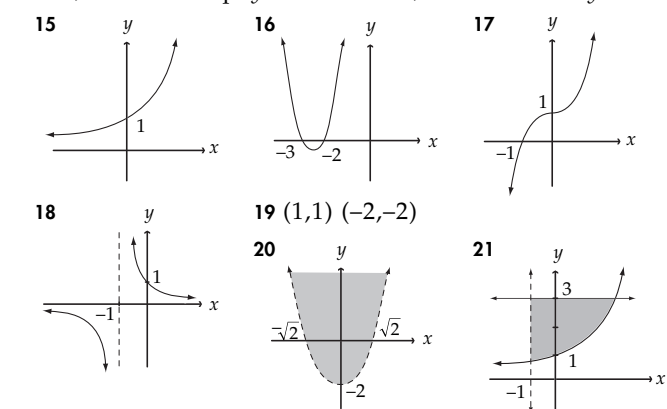


32 non-linear inequalities



33 functions: review 1 1 a 5 b -16 c -4 d $3x-1$
 2 a 20 b 15 c $3x^2+10x+7$ d $3(x+h)^2+4(x+h)$ 3 a 4 b 9 c $1\frac{1}{2}$
 d 13 4 a 28 b 1 c 1 d 344 5 a -9 b 9 c 6 6 even 7 odd
 8 even 9 even 10 all reals, all reals 11 all reals, all reals
 12 all reals except $x=-2$, all reals except $y=0$ 13 all reals
 except $x=-4$, all reals except $y=0$

34 functions: review 2 1 a 8 b -7 c 14 d -31 2 a 2 b 3
 c -6 d $2\frac{3}{4}$ 3 a 3 b 1 c $1\frac{1}{2}$ d $2\frac{1}{16}$ 4 a 29 b 47 c -7 d -22
 5 a 4 b 0 c -9 d $2\frac{1}{2}$ 6 a 11 b $\frac{1}{2}$ c 9 d 6 7 even 8 neither
 9 neither 10 odd 11 all reals, all reals where $y \geq 0$ 12 all
 reals where $-4 \leq x \leq 4$, all reals where $0 \leq y \leq 4$ 13 all reals except
 $x=0$, all reals except $y=0$ 14 all reals, all reals where $y \geq 0$



35 examination-style questions 1 1 c $D(1, 2\frac{1}{2})$,
 $E(3, -1\frac{1}{2})$ 2 a $C(2, 5)$ b 104° d $\frac{26}{\sqrt{10}}$ units e $2\sqrt{10}$

36 examination-style questions 2 1 a $x-y+3=0$
 b $4\sqrt{2}$ c $x-y-1=0$ d $x=5$ e $2\sqrt{2}$ f 8 square units
 2 a $D(1, -3)$ b $4x-7y-6=0$ d $\frac{19}{\sqrt{65}}$ e $9\frac{1}{2}$ square units,
 19 square units

37 examination-style questions 3 1 b 3 c 10
 d $\frac{22}{5}$ e 33 square units f 37° 2 a A $(-4, -\frac{1}{2})$ B $(-1, -2)$
 b $xy \geq 2$, $x+2y+5 \geq 0$ c $x=1$ y = $\frac{1}{2}$ d $\frac{\sqrt{45}}{2}$ f parallelogram
 g $\frac{10}{\sqrt{5}}$ units h 15 square units

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